THE NEO FAMILY OF COMPUTERS · TECHNICAL BOOK · VERSION 1.0

Neo inside

All adopted hardware and software decisions in one place, with chip specifications and a detailed concept of Neo OS. For those who want every register, slot and layer.



4 November 2026 · concept · all figures are engineering estimates, not measured values

0. Register of adopted decisions
1.1 Models and configurations
1.2 Chip specification
1.3 ROM, cartridge and compression
1.4 Slots and ports
1.5 Perception: Efficient and Pro
1.6 The Eye camera
1.7 Power and energy
1.8 Peripherals
2.1 Software layers
2.2 Living memory: no swap
2.2b Personality, everyday language and Dijana
2.2b2 Retro and Future Shock
2.2c Beauty without the burden
2.2d Sound as part of the experience
2.2e Co-op and the Board
2.2f Files without hassle and Cabinet Pro
2.2g Neo Workshop
2.2h Runs almost everything
2.2i The best of the Unix world
2.3 Apps in the box
2.4 Good old ideas
2.5 The Neo OS concept (in detail)
3.1 Path to a hardware standard
3.2 Prices and the path to production

PART 0

Register of adopted decisions

Where later chapters (especially the source documents in 1.2 and 2.4) differ from this table, this table applies.

Area	Adopted	Replaces
Main processor	A fast ARM is the sole master of the system and translates x86 programs (FEX/Box64 under Bridge)	Fluid CPU, several equal processors
Package	Neo package: ARM, FPGA/CGRA, VSA-6000, Neuron and other custom block chiplets, an optional x86 chiplet, UCIe link, HBM	Separate chips on the board
Memory	One coherent memory (CXL) for the ARM, FPGA, Satellite and all coprocessors, with QoS channels for picture and sound	Copying between memories
x86	Satellite: a replaceable x86 module (COM type) like the Bridgeboard. Absent in Neo 500, optional in 2000, built into 4000	Mandatory x86 in every model
FPGA	Chameleon loads retro cores (Z80, 6502, 68030, Amiga AGA) and accelerators. The CGRA merges part of Newton, Ray and the Orpheus DSP	Fixed retro chip
3D	VSA-6000: 2 to 8 identical chiplets, screen split into 64×64 tiles, T-Buffer 2 and Glide mode	One big GPU
OS in ROM	Neo ROM 2.0: 64 MB NOR (XIP kernel) + read-only pSLC A/B (2×16 or 2×32 GB). The SSD keeps only user data, mirrored	OS on the SSD
Memory	No swap to disk. In order: cache, the fast layer of RAD:, program code, compression in RAM, putting to sleep (Chronos), then a message. Documents and the fast layer survive a warm restart	Swap file
Personality and language	Easy mode uses everyday words (Desk, Folder, Bin, Workdesk, Caretaker…), Shelves instead of a Start menu, Helpers instead of Commodities, Dijana as a resident who doesn't impose, the Dijana's Day screen saver	Start menu, Clippy-style assistant
Beauty without the burden	Glass with contrast checking, Tiles as Panorama layers, Newton physics for windows and the Bin, Flip Through (3D overview with names, filtering and return to place), Calm/Lively/Lavish levels, animations up to 150 ms, Under the Hood and Show what I can do	Effects on the main processor, a 3D switcher without names or filters
Sound	Orpheus under 5 ms, sound from where events happen, music ducking, sound follows physics, a caption for every sound, sound themes in the Treasury like Residents (free Crystal, Wood, Chip, CC0), an open theme format	Fixed system sounds
Co-op and the Board	Power lending in a closed room with double checking, shared reproducible results, See how it works, clubs without a centre, the Board as a built-in BBS with Neighbourhood Mail, keys exchanged face to face, code sandboxed until two signatures	Central cloud and accounts
Files (Easy)	The Lisa approach: no Save and no mandatory name, Templates as a pad, a file in one place with links, views by time and person, piles, colour labels, Case, Bag for external disks	Folders, file names and manual saving
Ormar Pro	In the spirit of Directory Opus: multiple listers, button banks, actions per file type, Lux scripts, modules (SFTP, compare, batch rename, duplicates)	A simple two-panel manager
Neo Workshop	A Pro pack of free programs (Sculptor, Brush, Vector, Typeset, Darkroom, Montage Pro, Studio) once for €99 to 149, free for schools; half the revenue to the projects, changes back upstream	Subscription to closed packages

Runs almost everything	Double-click chooses the room: Bridge + Wine/Proton, full Windows (Satellite x86), Linux domain, Android room with microG, Chameleon/Arcade, web; the Does it work? database on the Board; kernel anti-cheat, macOS/iOS, Play Integrity apps and top-resolution streaming until DRM certification don't work	The user chooses the emulator or VM
Director	ArtCraft as an AI film studio in the Workshop: local models on Neuron, cloud only with consent, provenance label; licence to be checked	Closed AI services only
Two styles	Retro (a clean classic Mac look, no effects) and Future Shock (glass, tiles, physics, 3D); every Neo Kit app works in both without extra code, switching in one click, per-app exceptions, the Silver/Slate/Contrast/Phosphor colours apply in both, the Treasury checks both	One look for all
From the Unix world	Space bar preview, Services, Recipes, title icon, hot corners, Continue, Hand to a neighbour, One mouse, Moving house, Disk mode; Bonjour-style Neighbourhood, driverless printing, clones, dual packages, Magnifier; portals, firmware through Renewal, Neo in your pocket	Every feature as a separate program
Residents	Mascots of choice in the Treasury, the only commercial add-on: one-time purchase, most of the revenue to the author, well-known characters only under licence (first candidate: the snake from “Ide zmija”, DJ Nick DeeTronic), a resident sees neither files nor the network. Personal touches: name, accent colour, power-on sound, voice, Nook, anniversary	One fixed mascot
RAD: disk	A working disk in RAM that survives a warm restart, as on the Amiga, with Safe encryption and an optional copy to SSD at shutdown	Only a RAM: that gets erased
Media	Neo cartridge: NVMe over PCIe 4.0 ×2, 64 GB to 1 TB, about 3 GB/s, Safe signature, FRAM for saves. Optional UHD Blu-ray / M-DISC	Digital only
Compression	Open zstd and LZ4 in Courier, in both directions, for the whole system. EROFS image of the OS. Apps as a single neopak file	Our own algorithm, UPX per program
Perception	Efficient (default) and Pro, the hardware supports both. HAM modes dropped	HAM6/HAM8
OS kernel	Fortress: an seL4-class microkernel, services in Rust, Linux only as an isolated driver domain	Linux kernel as the base
Updating	Renewal: one signed A/B image (kernel, drivers, FPGA cores, Satellite firmware, microcode)	Separate drivers
Safety	Safe chip, MTE, IOMMU, encrypted RAM, physical switches for camera, microphone and network, Lock labels Personal/Work/Public	Software protection only
Backups	Guardian (Archive remains the database in Quartet)	The name Archive for both
Camera	Eye S/M/Pro included with every model, partner optics under licence, mechanical shutter	Camera as an add-on
Energy	100 W Sun panels in a chain, Sun station (MPPT + LiFePO4), 48 V DC input, Sunny mode	Built-in solar panel
Voice	Local: Whisper-class for dictation, Ask Neo, Piper-class for voice. Claude only as an option	Voice in the cloud
Retro games	Arcade: FPGA core, then emulator, then box, automatically. No third-party commercial ROMs	Manual choice of emulator
Knowledge	Knowledge cartridge 512 GB: Kiwix Wikipedia, OSM, Gutenberg, Kolibri and more, monthly refresh	Internet only
System sound	Neo sounds and theme, made by synthesis, CC0	
Mascot	Dijana, a basset hound with a partisan scarf	

Licences	OS under the GPL, hardware CERN-OHL-S, public sector without licence fees, commercial manufacturers pay for the name and certificate	Licence by political system

1.1 Models and configurations

	Neo 500	Neo 2000	Neo 4000
Case	In a keyboard	Desktop	Tower
Main CPU	ARM (Neo paket)	ARM (Neo paket)	ARM (Neo paket)
FPGA fabric	Medium, one core	Large	Large + CGRA
VSA-6000	2 chiplets	4 chiplets	4 to 8 chiplets
Satellite x86	None, translation only	Optional module	Built in
M.2	1	2	3
PCIe 5 / AGP-X	None	1 PCIe 5 + AGP-X	2 PCIe 5 + AGP-X
Optics	External, USB	Bay, optional	Bay, built in
Cartridge	Yes	Yes	Yes
DE-9	2 at the back	2 at the back	2 front + 2 back
Network	WiFi 7	WiFi 7 + 2.5G	WiFi 7 + 10G
OS in ROM	NOR + pSLC 2 × 16 GB	NOR + pSLC 2 × 32 GB	NOR + pSLC 2 × 32 GB
Safe chip	Yes	Yes	Yes
Privacy switches	Camera, microphone, network	Camera, microphone, network	Camera, microphone, network
Eye camera	Eye S, included	Eye M, included	Eye Pro, included
Independence	Built-in battery, 5G modem	5G modem, Sun station	5G modem, Sun station
HDMI input	None	Yes	Yes

> Memory per model from the chip specification: Neo 500 32 GB LPDDR5X, Neo 2000 64 GB LPDDR5X, Neo 4000 96 GB HBM3. The Neo 500 has no x86 part and translates x86 programs on the ARM.

1.2 Chip specification

> Source document from 3 November. Adopted later: ARM as the only main CPU, Satellite x86 as a module, a CGRA that merges part of Newton, Ray and the Orpheus DSP, and CXL for a coherent link with x86 and AGP-X cards.

2026-11-03

Target values for the final version on a 3 to 5 nm class process. All figures are engineering estimates from general knowledge, not measured or verified values, and serve to give the concept a realistic scope.

Package and models

All three models use the same chiplets; they differ only in the number of VSA-6000 chiplets, memory and power.

Parameter	Neo 500	Neo 2000	Neo 4000
CPU (ARMv9 or RISC-V RVA23)	8 cores	12 cores	16 cores
VSA-6000 chiplets	2	4	8
3D power (FP32)	~20 TFLOPS	~40 TFLOPS	~80 TFLOPS
Memory	32 GB LPDDR5X, 256 bita	64 GB LPDDR5X, 256 bita	96 GB HBM3, 3 stacks
Memory bandwidth	~550 GB/s	~550 GB/s	~2.4 TB/s
Shared cache	64 MB	96 MB	128 MB
Neuron (INT8)	100 TOPS	200 TOPS	200 TOPS
Display outputs	4	6	6
Power of the whole package	~150 W	~250 W	~450 W

Package: a base die with memory controllers, cache and bus, with chiplets above and beside it connected via UCIe. Newton, Ray, Panorama, Orpheus, Copernicus, Courier and Chronos are the same in all three models.

VSA-6000: the 3D chiplet

The name is a tribute to the Voodoo5 6000 card, on which four VSA-100 chips (Voodoo Scalable Architecture) drew one picture together. VSA-6000 is the same principle, except that the chiplets share memory instead of each having its own copy of the textures.

Parameter (per chiplet)	Value
Shader units	16 groups × 128 FP32 ALUs = 2048
Clock	~2.4 GHz
FP32 / FP16 power	~9.8 / ~19.6 TFLOPS
ROP units	32 (~77 Gpixels/s)
Texture units	64
Local tile SRAM	4 MB (64×64 pixel tile, 8 layers)
Support	Vulkan 1.4, mesh shaders, variable rate shading
Power	~35 W

Work distribution ("SLI 2.0", Screen-tile Link Interleave). A small hub on the base die receives the geometry, bins it by tile and hands tiles to the chiplets in a checkerboard pattern. A tile that is running late can “steal” a free chiplet, so none sits idle.

T-Buffer 2. The Voodoo5 had the T-Buffer: several samples of the same scene merged into one frame, for smooth edges, motion blur and depth of field. VSA-6000 brings it back as hardware accumulation: each chiplet can draw its own sample of the same tile (offset in time or position), and the hub blends them. So 4 chiplets give cinematic motion blur with no extra work for the programmer.

3dfx mode. A small compatibility layer for the Glide API, as a gift to the retro community: old Glide games run directly, without wrappers.

Newton and Ray

Newton: physics

Parameter	Value
Active rigid bodies	~500,000 at 120 Hz
Particles	~20 million per frame
Fluids (SPH)	~2 million particles
Broad-phase collision detection	Hardware BVH tree, refreshed every step
Narrow-phase collision detection	64 GJK/EPA units for convex shapes
Constraint solver	128 FP32 vector lanes, ~8 TFLOPS
Clock and power	~1.8 GHz, ~15 W

Determinism is achieved by a strict order of operations and IEEE rules without reordering computation steps, so the same input gives a bit-identical result on every unit. When a game doesn't use physics, Newton is available as general compute.

Ray: ray tracing, blitter and compositor

Parameter	Value
RT units	64, each with BVH traversal and ray-triangle intersection
Ray throughput	~20 billion rays/s (primary rays, simple scene)
BVH building	In hardware, for dynamic objects every frame
Blitter	64 pixels per clock, ~128 Gpixels/s fill rate
Blitter operations	Masked copy, fill, lines, Bézier curves, scaling, rotation, 12 Porter-Duff modes
Fonts	Hardware rendering via signed distance fields (SDF), sharp at every size
Compositor	Up to 64 layers per frame, 10 and 16 bits per channel
Audio ray tracing	Up to 64,000 rays per frame reserved for Orpheus
Power	~30 W

The blitter also has a “classic” mode with bitplane formats and minterm logic, so the demoscene can write effects the way they knew from the Amiga.

Panorama and Orpheus

Panorama: 2D and display

Parameter	Value
Independent outputs	6 (Neo 500: 4), each with its own resolution and refresh rate
Digital outputs	DisplayPort 2.1 (UHBR20), HDMI 2.1 (FRL 48 Gb/s), USB-C
Maximum per output	8K at 120 Hz with DSC, or 1440p at 500 Hz
Analogue output	RGB/VGA via a 10-bit 400 MHz RAMDAC, a tribute to Matrox quality, and a 15 kHz mode for CRT TVs and old monitors
Hardware layers	16 per output, each with scaling, colour conversion and transparency
Colour	10 and 12 bits, HDR10 and HLG, 33^3 3D LUT per output for calibration
Variable refresh	VRR from 1 to 500 Hz
Pull-down screens	Up to 8 zones per screen, each with its own resolution and palette, split on an exact scanline
Power	~5 W

Orpheus: sound

The digital part is a chiplet in the package; the analogue part is on the board, in a shielded zone with its own linear power supply and separate oscillators for the 44.1 and 48 kHz families.

Parameter	Value
DSP	8 cores at ~1 GHz
Voices	256, mixing in 32-bit float, 64-bit accumulator
Sound objects	Up to 128 positioned objects, output up to 7.1.4
Formats	Up to 32-bit / 384 kHz PCM, DSD256
Latency	1 ms buffer, guaranteed memory channel
Spatial audio	HRTF with personal profiles, head tracking via headphone sensors
Room correction	4096-tap FIR filter per channel, measured with a microphone
DAC (analogue part)	THD+N below -115 dB, dynamic range above 125 dB
Headphones	Amplifier for 16 to 600 ohms, ~500 mW into 32 ohms, 6.35 mm and balanced 4.4 mm outputs
Hi-Fi speakers	Balanced XLR line output (+4 dBu) and RCA
Digital outputs	S/PDIF optical and coaxial, HDMI eARC, USB audio
Bluetooth	5.4 LE Audio (LC3), aptX Lossless, LDAC, latency measurement for lip sync with the picture
Power	~4 W digital part

Retro mode: 4 "Paula" channels with 8-bit samples and the original filter, for accurate MOD music.

Copernicus: conductor of the system

The original Copper had only three instructions: WAIT, MOVE and SKIP. Copernicus keeps them as the core and adds what a system with nine chips needs.

Parameter	Value
Hardware threads	16, each with its own command list
Instructions	64-bit, fixed length
Global clock	1 GHz, 1 ns resolution, shared with all chips
Event response	Under 50 ns from event to register write
Event sources	Line and frame per monitor, end of job on each chip, audio buffer, input, timer, software signal
Power	under 1 W

Instruction	Meaning
WAIT event, mask	Waits for an event (e.g. line 540 on output 0)
MOVE register, value	Writes a value into a register of any chip
SKIP condition	Skips the next instruction if the condition is met
MOVEM address, count	Writes a block of registers from memory
KICK chip, job	Starts a job list on a chip (Newton, VSA-6000, Ray...)
SIGNAL event	Signals an event to other threads and the CPU
AT time	Waits for an exact moment of the global clock
JUMP, CALL, RET	Branching and subroutines
ADD, CMP	Minimal arithmetic on 16 registers per thread

Example: one thread runs the whole game frame, and the CPU only writes input and logic.

```
frame:
    WAIT   VBLANK(0)              ; start of a new frame on monitor 0
    KICK   NJUTN, physics_step    ; physics step
    WAIT   DONE(NJUTN)
    KICK   VSA, raster_tiles      ; VSA-6000 chiplets draw tiles
    KICK   ZRAK, rt_shadows       ; shadows and reflections in parallel
    WAIT   DONE(VSA) & DONE(ZRAK)
    KICK   NEURON, denoise_upscale
    WAIT   DONE(NEURON)
    KICK   ZRAK, composite        ; 3D + HUD + video
    WAIT   LINE(0, 1080)
    MOVE   PANORAMA.FB0, back_buf ; buffer swap exactly on the line
    JUMP   frame
```

A second thread can at the same time change the palette on certain lines for retro effects ("copper bars"), and a third can feed Orpheus.

Neuron, Courier, Chronos and the Retro FPGA block

Neuron: NPU

Parameter	Value
Compute power	100 TOPS INT8 (Neo 500), 200 TOPS (Neo 2000 and 4000)
Formats	INT4, INT8, FP8, FP16, BF16
Local SRAM	32 MB
Purpose	Upscaling, denoising for ray tracing, frame generation, character behaviour, voice commands, a local language model of 7 to 8 billion parameters (INT4)
Power	10 to 20 W

Courier: input, output and decompression

Parameter	Value
SSD	2 × NVMe over PCIe 5.0 x4
Decompression	Up to ~20 GB/s output: Zstandard, LZ4 and texture formats, directly into memory
DMA channels	32
Ports	4 × USB4, 10 Gb Ethernet, Wi-Fi 7, Bluetooth 5.4
Retro ports	2 × DE-9 for joystick and mouse (Amiga/Atari standard)
Input	A timestamp on every event at 1 µs, events go straight to Copernicus
Video	AV1, HEVC, H.264 encoding and decoding up to 8K at 60 Hz

Chronos: state snapshots and rewind

Parameter	Value
Change tracking	Per 4 KB page, in the memory controller
Snapshot per frame	Only changed pages, up to ~2 GB/s, into a 4 to 16 GB ring buffer
Rolling back	Up to 8 frames in under 1 ms (for network play with rollback)
Rewind	The last 30 to 60 seconds, depending on the game
Full system snapshot	To the SSD in under 1 s, including the registers of all chips

Chronos works only if the game is deterministic, which Newton and Copernicus guarantee for their part, while the programmer writes the CPU game logic with the help of an API for deterministic time.

Retro FPGA block

Parameter	Value
Capacity	~300,000 logic elements, 8 MB SRAM
Cores	Amiga OCS/ECS/AGA, C64 with SID, ZX Spectrum, Atari ST, arcade machines and consoles from the 16-bit era
Links	Video output as a layer in Panorama, sound as a voice in Orpheus, DE-9 ports via Courier
Power	2 to 5 W

This way the Neo-Amiga can also run original Amiga software cycle-exact, alongside modern games.

Motherboard: ports and slots

Courier provides all the lanes for the ports, and the most important one is the “AGP-X” slot: just as AGP in 1997 gave the graphics card direct access to system memory, AGP-X gives a card coherent access to the shared memory, as if it were one more chip in the package.

Port	Count	Specification	Purpose
AGP-X slot	1	PCIe 5.0 x16 + CXL 2.0 (coherent memory access, ~64 GB/s each way)	A stronger graphics card, a large FPGA, an AI accelerator
PCIe slots	2	PCIe 5.0 x4, open-ended (takes longer cards too)	Sound and network cards, video capture, retro cards
Satellite connector	1	Mezzanine, PCIe 5.0 x8 + CXL	Optional x86 Satellite module, takes neither the AGP-X nor the PCIe slots
M.2	3	2 × PCIe 5.0 x4 NVMe, 1 × PCIe 4.0 x4	Fast SSDs, the third for an extra disk or card
USB-C (back)	4	USB4 v2 / Thunderbolt 5, 80 Gb/s, power up to 100 W	Monitors, external disks, docking stations
USB-C (front)	2	USB 3.2, 20 Gb/s	Fast flash drives, controllers
USB-A	4	USB 3.2, 10 Gb/s	Keyboard, mouse, older devices
USB-B	2	USB 3.2, 10 Gb/s, device mode	The Neo-Amiga as an external Hi-Fi DAC or MIDI device for another computer
Wi-Fi	built in	Wi-Fi 7, 2×2, 320 MHz channels	Wireless networking
Bluetooth	built in	5.4 with LE Audio, together with the Wi-Fi module	Headphones, controllers
Ethernet	1	10 Gb/s	Wired networking
DE-9	2	Amiga/Atari standard	Joystick and mouse

About 40 PCIe 5.0 lanes from Courier in total. The USB-B ports are an unusual but useful choice: another computer or a phone can use Orpheus as a USB sound card.

Link with the CPU direction: the main ARM processor with x86 code translation stays in the package, the small Retro FPGA block stays for accurate emulation, and a stronger FPGA goes on a card in the AGP-X or a PCIe slot. The x86 Satellite has its own connector, so it doesn't take a slot.

New and future standards

The board is designed so that the next revision moves to newer standards without changing the chips in the package, because all the fast links go through Courier. The list is from my knowledge (up to mid-2026) and has not been checked against sources.

Area	In this version	Next step
PCIe slots	PCIe 5.0 (32 GT/s per lane)	PCIe 6.0 (64 GT/s) for AGP-X and M.2, PCIe 7.0 when it reaches the desktop
Coherent access	CXL 2.0	CXL 3.x, memory sharing with several cards at once

USB	USB4 v2 / Thunderbolt 5 (80 Gb/s, up to 120 Gb/s one way for monitors)	The successor to USB4 v2 once standardized
Monitors	DisplayPort 2.1, HDMI 2.1	HDMI 2.2 (96 Gb/s), already announced
Wi-Fi	Wi-Fi 7	Wi-Fi 8 (802.11bn), focus on reliability and lower latency
Bluetooth	5.4 with LE Audio	Bluetooth 6.x, precise distance measurement (Channel Sounding)
Memory cards	SD Express and CFexpress 4.0 reader on the front	Newer generations of the same standards
Network	10 Gb Ethernet	25 Gb Ethernet on the Neo 4000

Optical discs. For consumers there is no successor to Blu-ray; the industry has moved to streaming and SSDs. The proposal is an optional 5.25-inch bay for Ultra HD Blu-ray (100 GB, with M-DISC archival discs for long-term storage) or an external drive over USB-C. Archival formats such as Sony Optical Disc Archive and glass media (Microsoft Project Silica) exist, but they are for data centres, not home computers.

1.3 ROM, cartridge and compression

Neo ROM 2.0

- **NOR flash 64 MB**: execute in place (XIP), minimal kernel and recovery, like Kickstart.
- **pSLC flash A/B**: 2×16 GB (Neo 500) or 2×32 GB (2000, 4000), read-only. Contains the EROFS image of the OS and basic apps, compressed with LZ4/zstd (about 2× capacity).
- **Boot**: Chronos resume in a second or two, or a cold start loading the core (2 to 3 GB) into RAM in one go via Courier.
- **RAD: disk**: a working disk in RAM that survives a warm restart, a program crash, Renewal and a change of FPGA core. The zone is kept in self-refresh during reset, Neo ROM doesn't erase it, it is encrypted with a key from the Safe chip, with a checksum per block. Limit 4/16/32 GB per model, compressed via Courier. Option “Save RAD: at shutdown” (zstd to SSD); Chameleon cores can boot from RAD:.
- **SSD**: user data only, two M.2 drives mirrored. Compressed swap in RAM (zram) gives about 50% more space.
- **Updating**: Renewal writes to the inactive partition; an interrupted update doesn't leave the machine unable to boot. Service mode via USB-B.

The Neo cartridge and media

	Neo cartridge	UHD Blu-ray / M-DISC	Digital (Treasury)
Capacity	64 GB to 1 TB	50 to 100 GB	As much as there is room on the SSD
Reading	About 3 GB/s	About 100 MB/s	SSD speed
No installation	Yes	No, copied to the SSD	No, downloaded
Durability	Good, pSLC	M-DISC: archival, for decades	As long as you keep a copy
Ownership	Can be lent and sold	Can be lent and sold	DRM-free package, stored locally
Cost to the publisher	Highest	Low	Lowest
Best for	Games, cores, collector's editions	Films and archives	Everything else

- PCIe 4.0 ×2 NVMe interface, pSLC, signature via the Safe chip, FRAM for save states.
- Runs without installation, like a neopak package.
- Special cartridges: Knowledge (512 GB), Key (encrypted user profile), FPGA cores.

Compression

- Open zstd for almost everything, LZ4 when speed matters most. Courier does both in hardware, in both directions.
- Used in the file system, ROM image, neopak packages, Chronos snapshots, zram, network transfer, backups and cartridges.
- Every block carries an algorithm and version tag, so new algorithms are added through Renewal without losing readability.
- Cabinet opens ZIP, 7z and RAR, and creates 7z and zstd.

1.4 Slots and ports

Port	Description	Models
AGP-X	PCIe 5 ×16 + coherent CXL link + side channel for Panorama layers. Graphics and FPGA cards in the same memory	2000, 4000
PCIe 5	Ordinary expansion slots	2000: 1, 4000: 2
M.2	NVMe for user data, mirrored; 5G modem	500: 1, 2000: 2, 4000: 3
Cartridge slot	PCIe 4.0 ×2 NVMe	all
DE-9	Joysticks, like back then	500/2000: 2 at the back, 4000: +2 at the front
VGA	Quality RAMDAC from Panorama	all
HDMI / DP	Up to 6 outputs from Panorama	all
HDMI input	Capturing another machine, AV1 encoder	2000, 4000
USB-C / USB-A	Modern ports, power	all
USB-B service	Service mode and ROM recovery	all
DC 48 V	Directly from the Sun station	all
Network	WiFi 7; 2,5G (2000), 10G (4000); 5G modem	all
Satellite connector	COM-type module for x86	2000 (option), 4000

Security in hardware

- Safe chip: keys, disk encryption, qualified electronic signature.
- ARM MTE, an IOMMU for every device, encrypted RAM. Satellite and PCIe cards see only their assigned memory.
- Physical switches for the camera, microphone and network cut the power; software cannot turn off the LED next to them.
- Dual Neo ROM (A/B).

PART 1 · HARDWARE

1.5 Perception: Efficient and Pro

	Efficient (default)	Pro	Hardware
Screen	Renders at a lower resolution, and Neuron upscales to 4K. 8 bits with dithering.	Native resolution, 10-bit HDR, calibrated colours	Panorama scaler, Neuron
Textures in games	ASTC compression, 4 to 8 times less memory	Uncompressed or lightly compressed textures	VSA-6000 decoder
Shading	Variable rate: detail where the eye is looking	Every pixel at full quality	VSA-6000
Sound	Opus, the same as the original to the ear	Lossless FLAC, 24-bit	Orpheus decoder
Pictures	JPEG XL / AVIF, 5 to 10 times smaller	RAW and lossless JPEG XL	Courier codecs
Video	AV1	ProRes class for editing	Panorama encoder
Data	zstd everywhere	zstd everywhere (lossless in both modes)	Courier

On battery the system switches to Efficient by itself, except in apps marked as Pro. The original is always kept when it exists.

1.6 The Eye camera

	Eye S	Eye M	Eye Pro
Sensor	1/2"	1/1,3"	1", large pixels
Optics	Glass, fixed focus	Glass, autofocus, partner optics	Partner optics, bright, with a thread for swapping lenses
Video	4K 30	4K 60, HDR	4K 120, 10-bit log recording
Depth	Neuron only	Neuron only	ToF sensor + Neuron
Microphones	2	4, directional	4 + input for an external microphone
Connection	USB-C	USB-C	USB-C, raw stream directly into Panorama
Privacy	Mechanical shutter, an LED that software can't turn off, a physical switch on the computer case		

1.7 Power and energy

	Neo 500	Neo 2000	Neo 4000
Consumption	10 to 35 W	40 to 150 W	80 to 400 W
6 hours of use per day	About 150 Wh	About 500 Wh	About 1.2 kWh, about 600 Wh in power-saving mode
Panels in summer	1 × 100 W	3 × 100 W	6 × 100 W
Battery	Built-in 70 Wh + 300 Wh station	1 kWh station	2 kWh station
Without sun or mains	All day	About one day	One evening to one day

Other sources and saving

Generation	What it gives	Rating
Small 300–400 W wind turbine	Charges the station at night and in winter too, when there is no sun. A good complement to panels at a weekend cottage.	Worth it
Micro hydro on a stream	If there is a stream, it gives a constant 100 W or more, day and night. The best source where it is possible.	For special cases
Pedal or hand dynamo	A pedal gives 50 to 100 W while turning, and a hand crank 10 to 20 W. Enough for a Neo 500 in an emergency.	For special cases
Thermoelectric module on a stove	Turns the heat of a wood stove into 10 to 30 W. In winter it covers exactly the gap the panels leave.	For special cases
Piezo in the keyboard, radio waves	They give milliwatts, thousands of times too little for a computer.	Not worth it
Saving	**What it gives**	**Rating**
Sunny mode in Neo OS	Heavy jobs (rendering in Montage, Renewal, backups) wait for surplus sun, like a smart grid in miniature.	Worth it
Turning off chiplets	The Dispatcher fully turns off VSA-6000, the FPGA and Satellite when they aren't needed. On the desktop only the ARM runs.	Worth it
The right job on the right chip	Video encoding goes to Panorama, AI to Neuron, retro to the FPGA. A dedicated block uses ten times less than the CPU.	Worth it
Chronos instead of sleep mode	The computer turns off completely (0 W) and comes back in a second or two, so shutting down costs nothing.	Worth it
48 V DC input	Power comes straight from the station, without converting to 230 V and back. That's 10 to 15% less loss.	Worth it
Passive cooling	The Neo 500 has no fan. In the larger models the fans run only under load.	Worth it
E-paper monitor	For writing and reading in the sun, uses almost nothing. A nice option for the portable Neo 500.	For special cases
Heat as heating	All the power of a Neo 4000 ends up as heat. In winter the case can blow into the room, in summer outside.	For special cases

1.8 Peripherals

Device	How you get it	Description
Neo mouse	In the box	Three buttons, the middle one opens the menu
Small touchpad	In the box	Multi-finger gestures
Ergonomic keyboard	Extra cost	Split, mechanical, quiet switches, dual legends, Voice key
Neo remote	Extra cost	Arrows, numbers for teletext, air mouse, Voice button, sideways as a joystick
Neo microphone	Extra cost	Condenser, USB-C and XLR, mute button
Front panel	Built in	Volume dial, hardware microphone cut-off, Quiet, Voice, Chronos, Turbo and Retro, LED display

Every Neo also works with ordinary USB and Bluetooth peripherals, without drivers.

PART 2 · SOFTWARE

2.1 Software layers

Component	Role	Details
Fortress	Kernel	An seL4-class microkernel, formally verified. Linux as an isolated driver domain: when a driver crashes, the domain restarts.
Exec-N	IPC	Fast message and port passing, in the spirit of the Amiga Exec, in Rust.
Neo Kit	GUI and API	BeAPI spirit + Zune-style layout, Neo Platinum theme, Easy/Pro switch, the “Compress” call.
Compositor	Display	Wayland with the neo-menubar, neo-layer and neo-snapshot extensions. Panorama layers in hardware.
Lux	Scripts	ARexx-style ports in every app, automatically exposed as MCP tools too. Voice is translated into Lux.
Dispatcher	Scheduler	Chooses where a program runs: ARM, x86 translation, Satellite, FPGA or another Neo. Turns off chiplets that aren't needed.
Chameleon	FPGA service	Loads one retro core or accelerator at a time, signed through Renewal.
Bridge	Compatibility	FEX/Box64 for x86 on ARM, Wine and Proton for Windows programs and games.
Clasp	Windows	Windows and ReactOS windows behave like Neo windows, with shared clipboard.
Chronos	State	Hardware state snapshot: resume in a second, rewind, rollback netcode, system rollback.
Renewal	Updating	One signed A/B image of the whole system, rollback with one button.
Treasury	Programi	Open source, reproducible builds and a check that an installed program matches its code.
Lock + Safe	Privacy	Data labels, per-app firewall, keys in the Safe chip, capability-based sandbox (powerbox).
Datatypes	Formats	System add-ons for formats, multi-form clippings, live links.
Undo	History	Three levels: a step in a document, a file version, the system state.
neopak	Packages	Every app is one compressed, signed file that mounts without installation.
Guardian	Backups	Backups to disk, M-DISC or a cloud of your choice, at night or with surplus sun.
Arcade	Games	FPGA, emulator or box, automatically, with one menu for all.
Library and Ask Neo	Knowledge	Search of the Knowledge cartridge and a local model that answers with a source.

Voice

- Speech to text locally on Neuron, Serbian in both scripts, with punctuation.
- Commands are translated into Lux, so they work in every app.
- Ask Neo answers by voice (a natural Serbian voice, robotic as an option). Before an action that can't be undone it asks out loud.
- It listens only while the Voice button is held or after the wake word, if it is on.

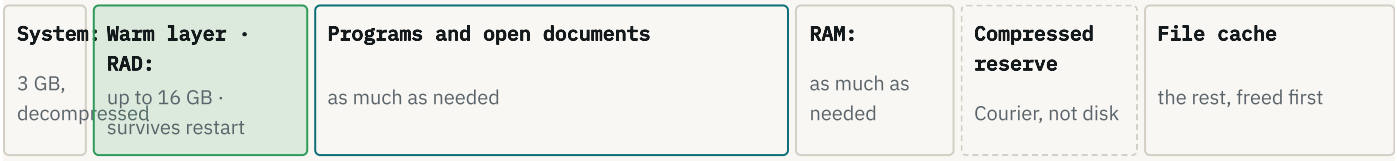
Arcade: order of choice

1. An FPGA core in Chameleon, if one exists (Amiga via Minimig-AGA and TG68, C64, Spectrum, NES, SNES, Mega Drive, old arcades).
2. Software emulator (MAME, libretro), with VSA-6000 and Neuron.
3. Box: Satellite x86, Wine/Proton, ReactOS.

PART 2 · SOFTWARE

2.2 Living memory: no swap

Neo never moves memory to disk. RAM:, RAD: and programs share one compressed space. Open documents and everything important for speed survive a warm restart. Sizes are a proposal for a Neo 2000 with 64 GB.



The limits aren't fixed: while RAM: and RAD: aren't using space, it serves programs and the cache. The warm layer is marked in green because the memory controller preserves it during a warm restart.

When memory runs out, in order	What happens	The user notices
1. File cache	Freed first, because it can be read again.	Nothing
2. The fast layer of RAD:	The part of RAD: with pre-prepared programs and libraries shrinks. Live copies of open documents are never touched.	Nothing
3. Program code	Programs from ROM and neopak packages are read-only, so their code is released and reloaded when needed. No swap is needed for them.	Nothing
4. Compression	Rarely used memory is compressed in RAM via Courier, in hardware, about 2 to 3 times smaller.	Almost nothing
5. Putting to sleep	The program you haven't used the longest is frozen, and Chronos writes a compressed snapshot of its whole state into the warm layer. A click wakes it in a fraction of a second, in the same state.	The program icon gets a “sleeping” badge
6. Asking	If even that isn't enough, Neo says which programs take up the most and offers to close some. It never slows down the whole system.	One message, instead of freezing

Warm layer

What survives a warm restart

DOCUMENTS

Every open document, with unsaved changes, undo history and cursor position. Every change is immediately written to a journal in RAD:.

PROGRAMI

The state of open programs and windows, the clipping Collection and the loaded FPGA core.

SPEED

The decompressed system core, the shader and font cache, the search index, shared libraries and the list of programs you use most, already prepared.

Durable

RAM speed, disk safety

WRITING

Every change goes to RAD: immediately, and Courier copies it compressed to the SSD within a couple of seconds. Saving is therefore instant.

POWER LOSS

The Neo 500 has a battery. The Neo 2000 and 4000 get a supercapacitor in the power supply, which gives a second or two to flush the last writes to the SSD.

COLD BOOT

Courier restores the warm layer from the SSD with hardware decompression, in two to three seconds for a few gigabytes.

Dynamic

RAM: and RAD: grow and shrink

RAM:

Temporary, erased at restart. Takes up only as much as is in it.

RAD:

Survives a restart, has a permanent copy on the SSD and a limit per model: 4, 16 or 32 GB.

FAST LAYER

The Dispatcher tracks what you use and keeps the most used programs and libraries in the fast layer of RAD:. Films and large archives stay on the SSD.

Hardware

What the machine must have

CONTROLLER

A memory zone in self-refresh during a warm restart, which Neo ROM doesn't erase.

INTEGRITY

ECC and periodic memory scrubbing, because data lives in RAM for a long time. A per-block checksum in the warm layer.

PROTECTION

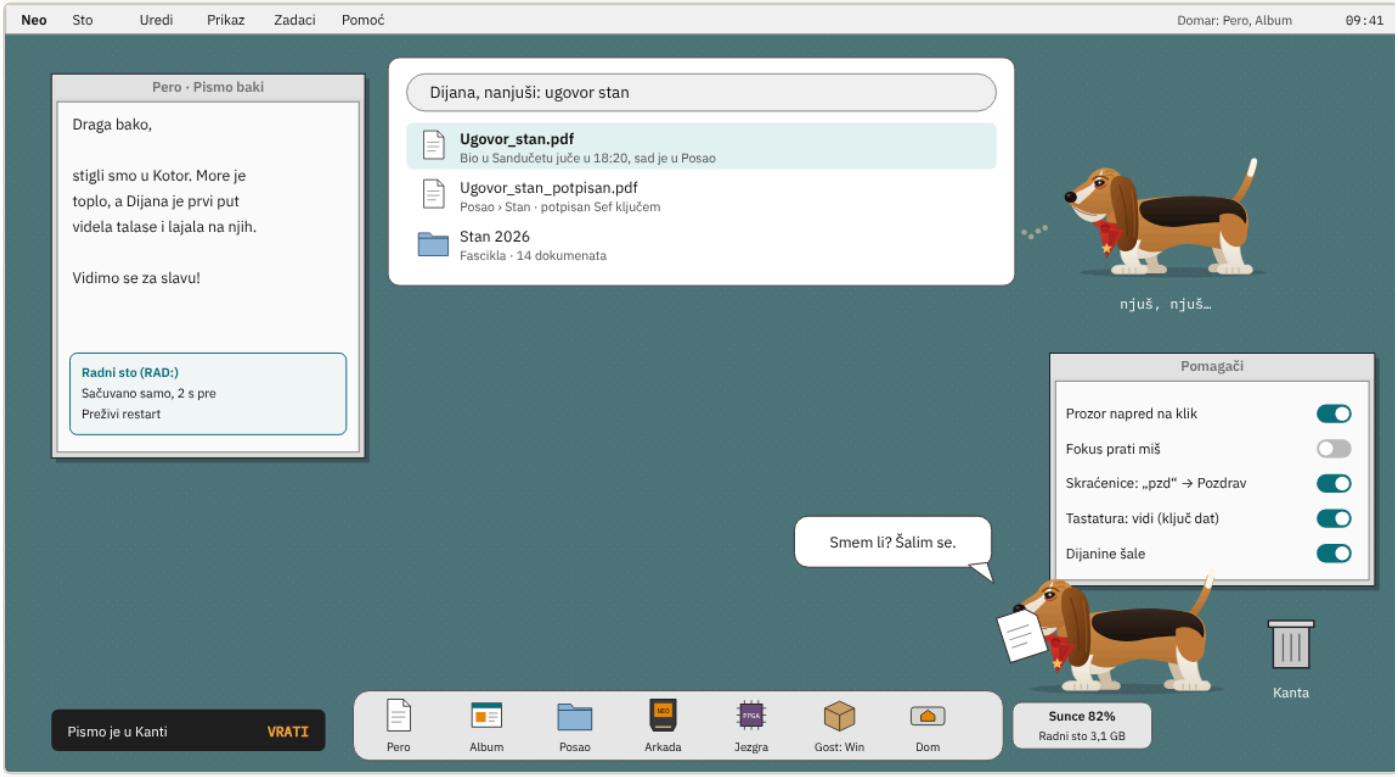
The warm layer is encrypted with a key from the Safe chip. Network keys and passwords aren't kept and are asked for again.

> Estimate, not measured: first program launch from about 300 ms to under 100 ms, return after a warm restart from about 10 s to about 2 s, with all windows and documents in place. In already warmed-up work the gain is 10 to 20%, but there are none of the stalls swap causes when memory fills up, and saving is instant.

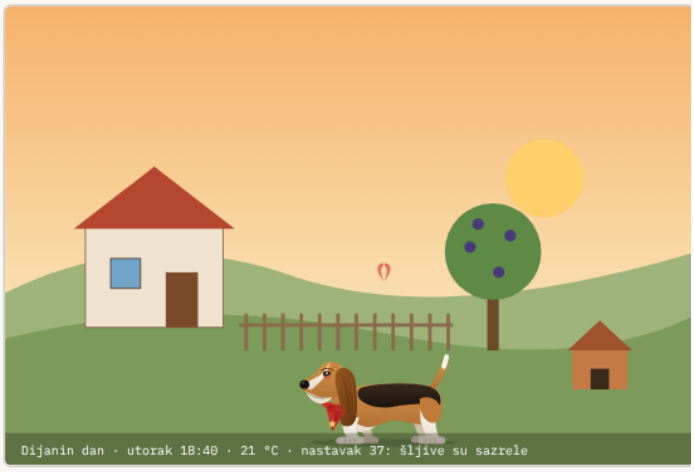
PART 2 · SOFTWARE

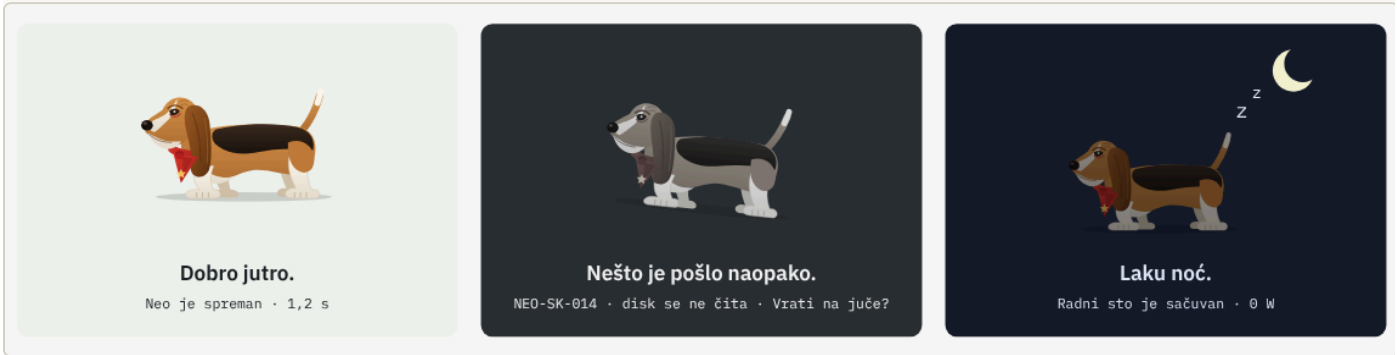
2.2b Personality, everyday language and Dijana

Everyday words in Easy mode (technical terms in Pro), shelves instead of a Start menu, Helpers, Dijana as resident and guide, the Dijana's Day screen saver and simplicity modelled on good mobile apps. The name Blocks: is called Templates in Easy mode.









Easy (everyday)	Pro (technical)	Why
Desk	Desktop	On the desk is what you are working on right now.

Easy (everyday)	Pro (technical)	Why
Folder	Folder, directory	Just like a paper folder.
Mailbox	Downloads	Everything that comes from outside arrives here.
Bin	Trash	Can be brought back until it is emptied.
Foundation	System:	Everything stands on it and nobody touches it.
Toolshed	Apps:	All programs in one place.
Guest rooms	Boxes:	Windows and other guests live separately.
Notepad	RAM:	Quick notes that get erased.
Templates	Blocks:	Ready-made beginnings for a letter, an invoice or a project.
Workdesk	RAD:	Stays the way you left it, even after a restart.
Caretaker	Dispatcher	Knows who does what in the house and what uses power.
Keys	Permissions	Who may enter which room.
Neighbourhood / World	Local network / internet	Nearby computers and everything else.
Household / Kids' room	Profiles / child profile	Everyone has their own place.
Back to yesterday	Chronos rollback	Says exactly what it does.
Nap / Good night	Sleep / shutdown	Just like people.
Shelves	Dock, launcher	You put on the shelf what you need at hand.
Helpers	Background services	Little helpers, each for one job.

Dijana

A living resident, not a pushy assistant

WHERE SHE LIVES

She sleeps in the corner of the menu bar. When she has a useful tip, she just raises her ears. Click if you want, otherwise she keeps sleeping.

WHEN SHE SPEAKS UP

At first power-on, after an error (to explain what happened) and when you call her by voice or with a key. Never in the middle of typing.

LEARNS

Every tip only once. “I don't need this” turns off that whole kind of tip, and after being ignored three times she goes quiet on her own. She never sells anything.

Sniffing

“Dijana, sniff out…”

SEARCH

A basset is a scent hound, so search sniffs: Dijana lowers her nose and follows the trail, and the results come up as you type.

LOST

For a lost file she says where and when it was last seen: “It was in the Mailbox yesterday at 18:20, then you moved it to Work.”

Humour

Trying to eat a document

WHEN

Rarely: when you drag a document to the Bin, she sometimes catches it in her teeth and asks “May I?”, then drops it and wags her tail.

LIMITS

She never really touches the document, never in the middle of work and never more than once a day. Jokes are turned off with one switch.

On the desk

A small presence that doesn't get in the way

BEHAVIOUR

Sometimes she walks along the shelf, lies down next to a program that runs for a long time and wags her tail when Guardian has finished a backup. When the Sun battery runs low, she lies down in the shade.

PEACE

While you type or watch a film, she doesn't move. On battery she isn't animated, so as not to use power.

Screen saver

Dijana's Day

WHAT IT IS

A living story in episodes: Dijana in the backyard, at the market and by the river. Time of day and weather follow the real weather outside, and holidays and your calendar change the scene.

EPISODES

The story continues where it stopped. It is assembled from pieces on Neuron, so it rarely repeats. Our own original story and drawing.

System personality

Like the smiling Mac, only ours

STARTING UP

Dijana wakes up, stretches and raises her ears: everything is fine.

ERROR

Sad Dijana with lowered ears, an error code and the solution in plain language.

SHUTDOWN

“Good night”, and Dijana curls up. Along with soft sounds from Neo sound and animations that can be turned off.

Residents

Mascots of your choice, the only shop in the system

WHAT THEY ARE

A resident replaces or complements Dijana: a panda, bear, snake, cat, hedgehog, stork or a cartoon character. Each has the same behaviour (sleeps, sniffs during search, jokes, rejoices and mourns), its own screen saver story and its own sounds.

BUYING

In the Treasury, once and forever, without a subscription. The author gets most of the revenue, so local illustrators and animators get a market. Free community residents come under a free licence.

LICENCES

Well-known characters only with the official licence of the rights holder (cartoons, comics, games). First candidate for a deal: the snake from the song “Ide zmija” (“Here Comes the Snake”) by DJ Nick DeeTronic, as a dancing resident that dances to the Turntable. No unofficial copies and no real people. A country can add a national character, for example an animal or a fairy-tale character.

SAFETY

A resident only receives events (“search started”, “backup done”). It sees neither files nor the network, sells nothing and shows no ads.

Personal

A computer that is only yours

NAME AND GREETING

The computer gets a name, and the resident greets you by yours.

LOOK

One accent colour throughout the system, and the background. Neo checks the contrast, so everything stays readable.

SOUND AND VOICE

You choose the power-on sound or compose your own in the Orpheus synthesizer. You choose the voice and how much Neo talks.

NOOK

A small place on the desk for a picture, a quote or the Flowerpot, a plant that grows as you use the computer and never wilts.

ANNIVERSARY

“Today it's been a year since we've been together”, and a short look back at what you made. Everything stays on the computer.

Shelves

Instead of a Start menu

HOW

As many shelves as you want, along any edge. Anything goes on them: programs, documents, folders, Lux scripts and live tiles (clock, mixer, Workdesk, Sun).

SEPARATE

Shelves are for launching, and what is running is in the Caretaker in the menu bar. Each task has its own shelves. For everything else there is the command palette.

Helpers

Instead of Commodities

WHAT THEY ARE

Small background tools: hotkeys, bring window to front on click, focus follows mouse, abbreviation expansion, automatic actions.

SAFE

One panel turns them on and off. A helper that sees the keyboard or mouse must request it through Keys.

Like a phone

As simple as a good mobile app

SCREEN

One main action per screen, a big button in the same place. Search is always at the top, and Easy/Pro hides the rest.

NO FEAR

Instead of “Are you sure?” the action is carried out, and “Undo” waits at the bottom. There is no Save button, everything saves itself.

EMPTY

An empty screen isn't empty: it says what is done here and offers a “Start from…” template.

Task-based

“What are you doing today?”

GETTING STARTED

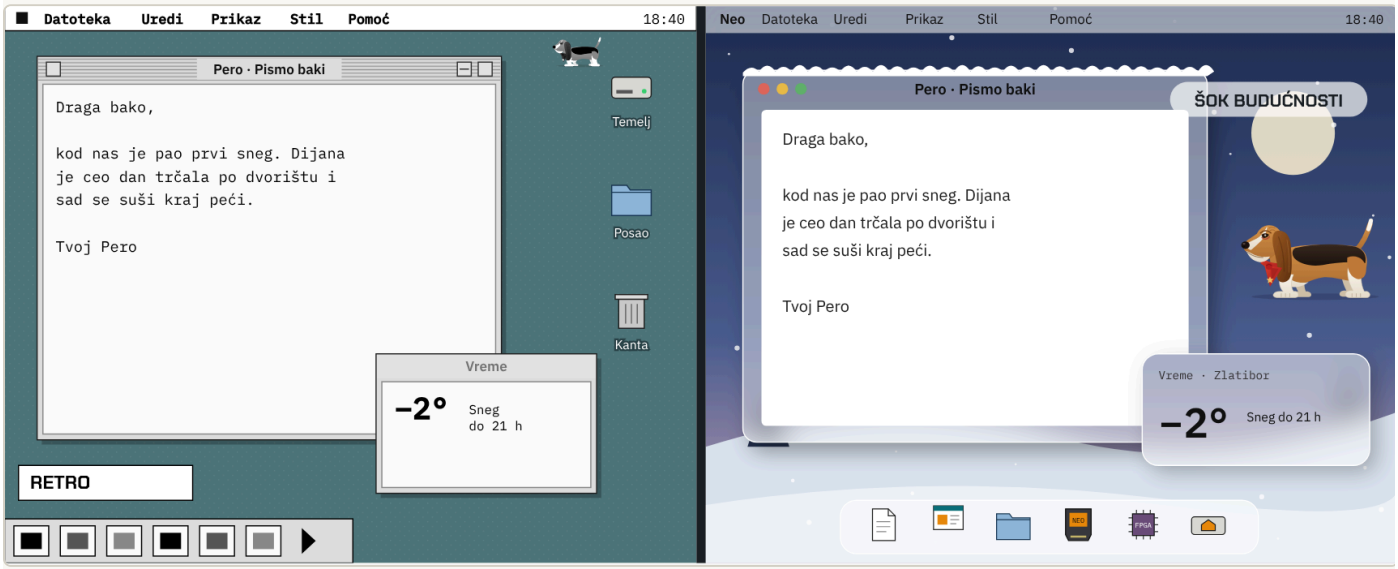
The first screen offers big tiles: Write a letter, Look at pictures, Call someone, Play, Learn. Each one opens a ready-made task with the right programs and shelf.

LEARNING

First steps through a game in 15 minutes, and every program shows three things the first time, no more.

2.2b2 Retro and Future Shock

Neo has two global styles. Retro is a clean look in the spirit of the classic Mac systems of the nineties: grey silver, striped title bars, crisp letters and not a single effect. Future Shock is everything from the Beauty section without the burden: glass, tiles, physics, 3D and the seasons. Every app runs in both without a single extra line of code, and switching takes one click.



Part of the system	Retro	Future Shock
Window	Grey frame, striped title bar, square buttons, hard shadow	Frosted glass, rounded corners, soft shadow following the time of day
Menu	White bar with black letters, drop-down menus without transparency	Glass menu with the beauty level and a clock
Shelves	Control strip along the edge of the screen, icons without motion	Glass shelf, icons bounce, snow settles
Window overview	List of windows in the application menu	Flip Through in a 3D arc
Tiles	Small add-ons in their own little window, like desk accessories of old	Live glass tiles
Bin	The Bin bulges when it is full	The document crumples up and flies off in an arc
Type	Cube, our crisp screen typeface, at size 12	IBM Plex Sans with smooth edges
Icons	32 × 32, limited palette, black-and-white if you like	Full colour with depth and shadow
Sound	Basic or Chip theme	Crystal or Wood theme, with spatial sound
Dijana	A small black-and-white pixel Dijana in the corner of the menu	Full Dijana who walks around, shakes off snow and chases paper
Motion	Just an outline that expands when a window opens	Physics, bounce and motion blur

One codebase

The app says what, the style says how

NEO KIT

The app describes a button, list, field and toolbar by meaning, not by appearance. Panorama draws the style, so the same app gets both faces without changes.

ITS OWN DRAWING

Apps that draw themselves (games, Sculptor, Arcade) keep their own look, and the frame around them follows the style.

GUESTS

Windows, Linux and Android programs get a frame and menu in the chosen style, and stay as they are inside.

Switching

One click, no restart

INSTANTLY

The style changes with a shortcut or from the menu, instantly in all open windows, without losing any work.

PER APP

Each app can have an exception: for example Cabinet in the Retro style and the rest in Future Shock.

POWER

On battery Neo suggests Retro with a single message, because Retro uses almost nothing for graphics.

Combinations

Style, colours and level are separate

COLOURS

Silver, Slate, Contrast and Phosphor work in both styles. Retro with Phosphor looks like an old green monitor, and Future Shock with Slate like a city at night.

LEVELS

Calm, Lively and Lavish apply within Future Shock. Retro is always calm.

ACCESSIBILITY

Contrast, magnification and “Less motion” work the same in both styles.

For developers

Checking in both styles is mandatory

TOOL

Neo Studio shows the app window in both styles side by side while you build it, and warns when something doesn't fit or loses contrast.

TREASURY

An app passes an automatic check in both styles and both colour schemes before publication. Whoever draws their own controls must provide both versions.

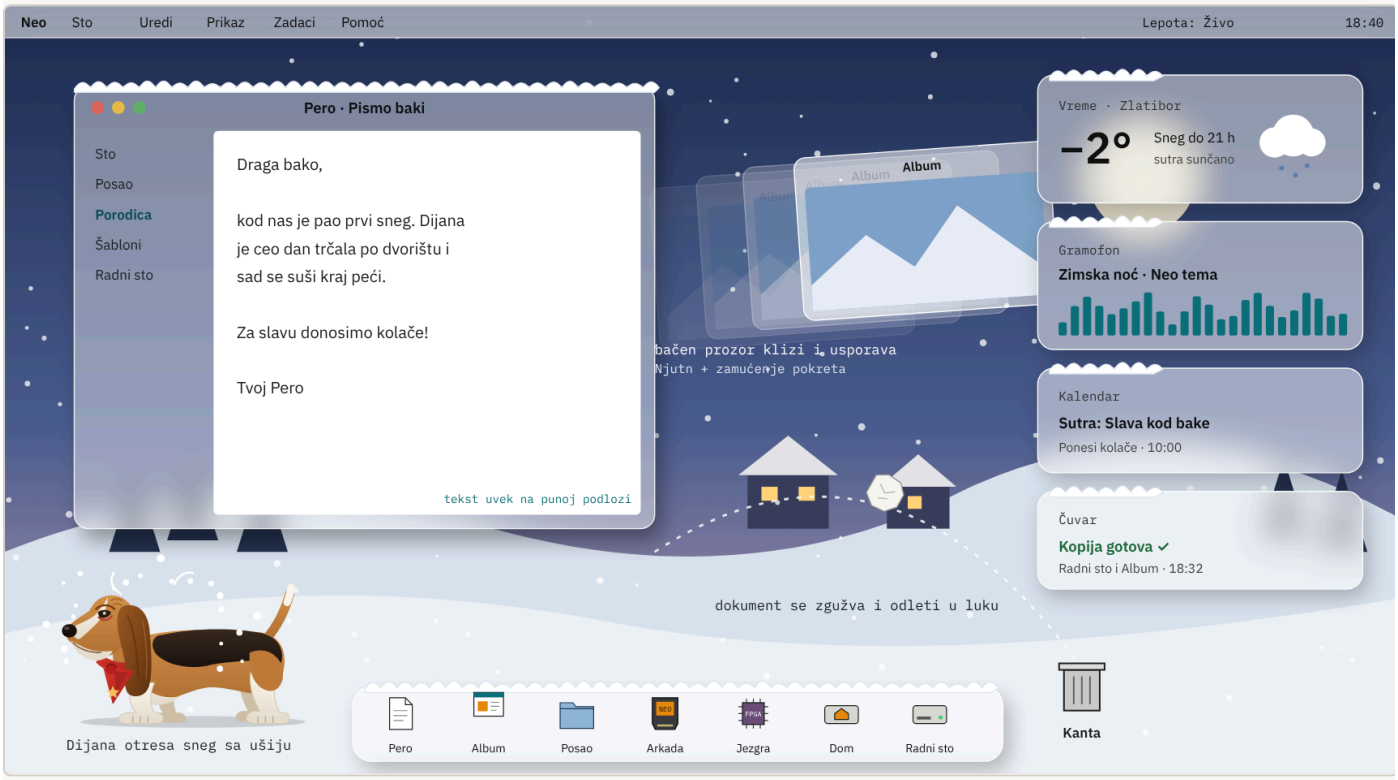
NAME

“Future Shock” is a nod to the famous book about how people experience rapid change. That is exactly why Retro exists too: whoever doesn't want change doesn't have to put up with it.

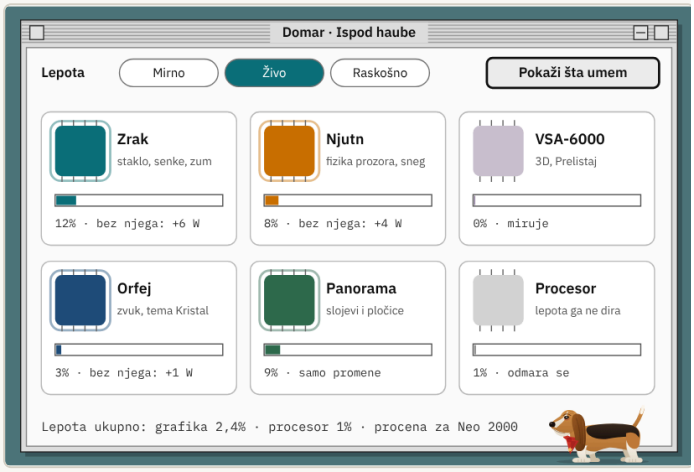
PART 2 · SOFTWARE

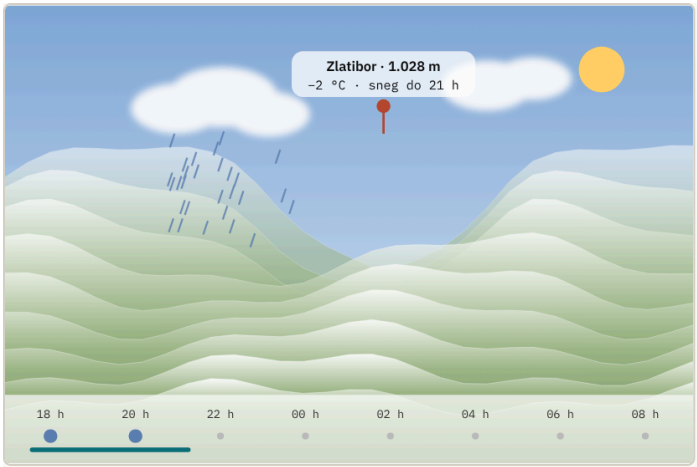
2.2c Beauty without the burden

The best from the era when computers loved to show off: tiles with live data, transparent glass, a 3D window overview and physics like in games. The difference is that on Neo all of this is done by dedicated chips (Ray for 2D and glass, Newton for physics, VSA-6000 for 3D, Panorama for screen layers), so the main processor and the battery barely feel it. Nothing slows down work, and everything turns off with one switch.











Tiles

Live data on the desk, the shelf or in the Nook

WHAT THEY ARE

Small live cards: weather, calendar, Turntable, exchange rate, Sun station, Guardian backups, a note, a clock for another city. The idea of the Vista-era sidebar, but without the slowness.

HOW THEY WORK

Each tile is a separate layer in Panorama and refreshes only when its data changes, at most once a second. When you can't see it, it does nothing.

SAFETY

A tile gets only the data it displays, through Keys. It cannot read files or send data without permission. They are made in Neo Kit in an afternoon.

Glass

Transparent, but always readable

WHERE

The menu bar, window titles, sidebars, Shelves and tiles are made of frosted glass through which the background shows. Text and documents are always on a solid, opaque backing.

CONTRAST

Ray measures the background under the glass and increases the blur or darkening by itself, so text never drops below the required contrast. With the Contrast theme the glass becomes solid.

LIGHT

Window shadows follow the time of day: short and cool in the morning, long and warm in the evening. The reflection on the glass shifts slightly when you move a window.

Newton

Physics that helps, not just entertains

WINDOWS

A thrown window glides and slows down, then bounces off the edge of the screen or moves to another monitor. Windows spread apart slightly so they don't overlap when there are many.

BIN

A document crumples up and flies in an arc into the Bin. The Bin fills visibly, and emptying is a short shake.

SHELVES

Icons on the shelves move aside when you drag something, and a program that is starting bounces gently. A folder dragged over another folder springs open.

SEASONS

In winter snow falls and settles on window edges and the shelf, in autumn leaves, in summer plums from Dijana's backyard. Move a window and the snow falls off. Only at the Lavish level, and never over a document.

VSA-6000

3D capabilities in everyday work

PRELISTAJ

In one move all windows fit into a 3D arc, with depth of field: the active one is sharp, the others slightly blurred.

ACCUMULATION EFFECTS

The spirit of the old T-buffer: motion blur on a thrown window, soft shadows, depth of field and perfectly smooth letter edges, all in one pass on the GPU.

ZOOM

Maps, pictures and PDFs zoom smoothly and without blur, because they are drawn from vectors and tiles at full resolution. The magnifier for the visually impaired works the same way.

RETRO

Glide mode and the Voodoo core in Arcade for old 3D games, with a sharper image than back then.

Every day

Where 3D helps an ordinary user

FORECAST

Weather as a live scene: clouds, rain and sun over the Atlas terrain of your area, with a clock across the whole day. The same scene can also be the desk background.

SOUND

Turntable shows music and streams as a gentle visualization, and an audiobook as a voice waveform with chapters. A calm story, a calm picture.

PICTURES AND VIDEO

Album as a light table with pictures that stack and turn. Video calls with a blurred background and light correction on Neuron.

READING

Leaf turns a page like paper, and a comic leads you panel by panel.

Revived

Apps that others threw out but people loved

GAMES

Cosmos Pinball with real ball physics, Grandmaster 3D with glass pieces, Solitaire and Minesweeper with cards that cascade at the end of the game.

TOOLS

Simple film editing from Album (title, music, transitions) in five minutes, a live background (Dijana's Day on the desk) and tiles.

RULE

They are all our original apps with our own names. We take the idea, not the code or the look.

Residents

Physics and residents together

DIJANA

She shakes snow off her ears, runs after a crumpled paper and brings it back, and lies down on the Sun station tile when it's warm.

OTHERS

The panda catches falling leaves, and the snake dances to the rhythm of the music from Turntable, because Orpheus detects the beat.

LIMIT

A resident gets only physics and beat events. It never covers a document and doesn't move while you type.

Flip Through

A 3D switcher that is also practical

LESSON

The old 3D switcher with windows stacked like cards looked great, but the windows at the back were hidden and had no names. Neo takes only what helps.

LAYOUT

Windows fit into an arc, all visible and named, grouped by Tasks. They stay live: video plays, a copy progresses.

FAST

Typing filters (“let” leaves Letter), the chosen window flies exactly into its place so the eye doesn't lose track, and it all takes under 150 ms. It works across all monitors, and the ordinary window list stays for those who like it.

DEPTH ON A FLAT SCREEN

Good 3D on a 2D surface: slight background parallax when the mouse moves, soft shadows that show what is on top and layered glass. Depth of field only in Flip Through.

Show what I can do

Hardware power anyone can understand

UNDER THE HOOD

A window in the Caretaker with a picture of the chips that light up while they work: Ray draws glass, Newton computes snow, Orpheus plays, VSA-6000 drives 3D. The main processor shows that it is resting, and next to each chip it says what would drain the battery without it.

DEMO PROGRAM

The “Show what I can do” button plays 30 seconds of a demo in the spirit of the demoscene, in real time: a flyover of Atlas terrain, a hundred windows with live video in Flip Through, thousands of snowflakes and Dijana. At the end, one sentence: “This ran while the processor was almost idle.”

SHOP WINDOW

A mode for shops and fairs that loops the demo and invites a passer-by to move a window or throw a document into the Bin.

Rules

Beautiful, but not a burden

LEVELS

Calm: no motion and no glass, for work and weaker eyes. Lively (default): glass, tiles, short animations and window physics. Lavish: everything, plus seasons, the 3D overview and a live background.

SPEED

No animation lasts longer than 150 ms and each one is interrupted by the next click. Work never waits for an effect.

POWER

On battery and in Efficient mode the system lowers the level by one by itself. Estimate for Neo 2000: Lively takes under 3% of the GPU, Lavish under 10%, and the main processor almost nothing.

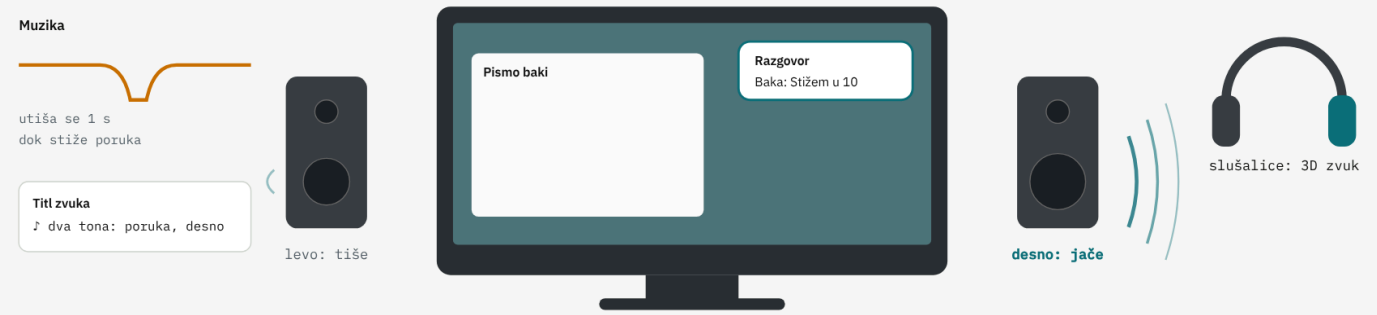
ACCESSIBILITY

“Less motion” and “No transparency” apply across the whole system and in all apps from Neo Kit.

PART 2 · SOFTWARE

2.2d Sound as part of the experience

Good sound tells you what happened before you look: it arrived, it worked, it failed, it's done. Neo sounds are handled by the Orpheus chip, with latency under 5 ms, so the sound arrives exactly with the motion on screen. The rule is the same as for beauty: useful, short and quiet, and every sound also has a caption.



Example sound themes made by synthesis, CC0, free with the system

Theme	Files in neo/zvuk/teme/ (theme name, hyphen, event)					
Crystal glass bells, for Glass	paljenje.mp3 (power-on)	otvori.mp3 (open)	obavestenje.mp3 (notification)	greska.mp3 (error)	kanta.mp3 (bin)	gotovo.mp3 (done)
Wood marimba, warm and quiet	paljenje.mp3 (power-on)	otvori.mp3 (open)	obavestenje.mp3 (notification)	greska.mp3 (error)	kanta.mp3 (bin)	gotovo.mp3 (done)
Chip 8-bit, like the home computers of the 80s	paljenje.mp3 (power-on)	otvori.mp3 (open)	obavestenje.mp3 (notification)	greska.mp3 (error)	kanta.mp3 (bin)	gotovo.mp3 (done)

Meaning

Every sound says something

LANGUAGE

Success goes up, an error goes down and short, a notification is two tones, and a finished long job is a small chord. The same rule applies in every theme, so the meaning is learned once.

MEASURE

Frequent actions (click, open) last under 150 ms and are barely audible. Longer sounds only for rare moments: power-on, a finished backup, an error.

HARMONIOUS

All sounds of one theme are in the same key, so even when two overlap they sound like a chord, not noise.

Space

Sound from where the event happens

PANORAMA

A notification from a window on the right is heard on the right, and from another monitor from that side. On headphones Orpheus does it in 3D.

DUCKING

When a message arrives or the Neo voice speaks, music and films dip for a moment, then come back.

RHYTHM

System sounds don't pile up: ten notifications in a row are heard as one, with a number on screen.

Physics

Sound follows Newton

MOTION

A thrown window rustles quietly, louder the faster it is. A crumpled document rustles in flight and lands in the Bin with a dull thud, and a bigger file sounds deeper.

PROGRESS

Optionally, a long copy has a quiet tone that rises as the job progresses, so you know how far it has got without looking.

GAMES

Pinball, cards and pieces get their sound from the same physics, not from recorded samples.

Dijana

A resident you hear, but who doesn't talk

SOUNDS

Sniffing while searching, a yawn when waking up, a short “woof” when the backup is done and tail thumping. No words and no tiresome sentences.

HOT, COLD

During a search the sniffing gets faster and higher as it nears the right result, like the game “hot and cold”.

Ambience

A soundscape if you want it

DAY

A quiet background that follows Dijana's Day and the real weather outside: rain when it rains, birds in the morning, crickets in the evening. Off until you turn it on.

FOCUS

Noise for concentration, a café in the distance or a library. A work and break timer with a soft bell.

Accessibility

Sound that guides

SCREEN READER

Short tones for a window edge, the end of a list and a button that can't be pressed, so a blind person hears the shape of the screen.

CHARTS

Table and Atlas can “play” a chart or terrain height, with a tone that rises and falls.

CAPTION

Every sound has a text and visual counterpart, for those who can't hear.

Silence

Silence is a feature

BUTTON

The Quiet button on the case and keyboard turns off all system sounds, and the music stays.

NIGHT

After 10 pm sounds are automatically quieter and softer, and during a call in Chat the system stays silent.

Time of day

Sound that follows the day

IDEA

Optionally, the theme changes colour through the day: wood and brighter tones in the morning, softer and lower glass in the evening. The meaning of the sounds stays the same.

STARTING UP

The power-on sound changes slightly with the seasons and holidays, and your own one from the Orpheus synthesizer stays as it is.

Retro

Real sound chips in Orpheus

CORES

Orpheus can load FPGA cores of classic home computer sound chips, so the Chip theme and retro games sound like they used to, with real noise and filter.

FOR AUTHORS

A theme can also be written as a small music file for such a chip, a few kilobytes in size.

Sound themes

Themes in the Treasury, like Residents and alongside them

WHAT A THEME IS

A pack of up to 40 sounds for system events, plus power-on and shutdown sounds and an optional ambience. Free with the system: Neo Basic, Crystal, Wood and Chip.

BUYING

In the Treasury, once and forever, without a subscription. The author gets most of the revenue, so local composers and sound designers get a market. Community themes come free under a free licence.

WITH A RESIDENT

Every resident can bring its own theme, and they are also sold separately. The dancing snake resident from “Ide zmija” gets a theme with parts of the song for power-on and a finished job, but only with the official licence of DJ Nick DeeTronic.

MIXING

Each event can come from a different theme: power-on from Chip, the Bin from Wood. An app can have its own theme, for example Arcade.

OPEN FORMAT

A theme is Opus or FLAC files and one text list of events. It is made in Wave or the Orpheus synthesizer and shared as a single neopak file.

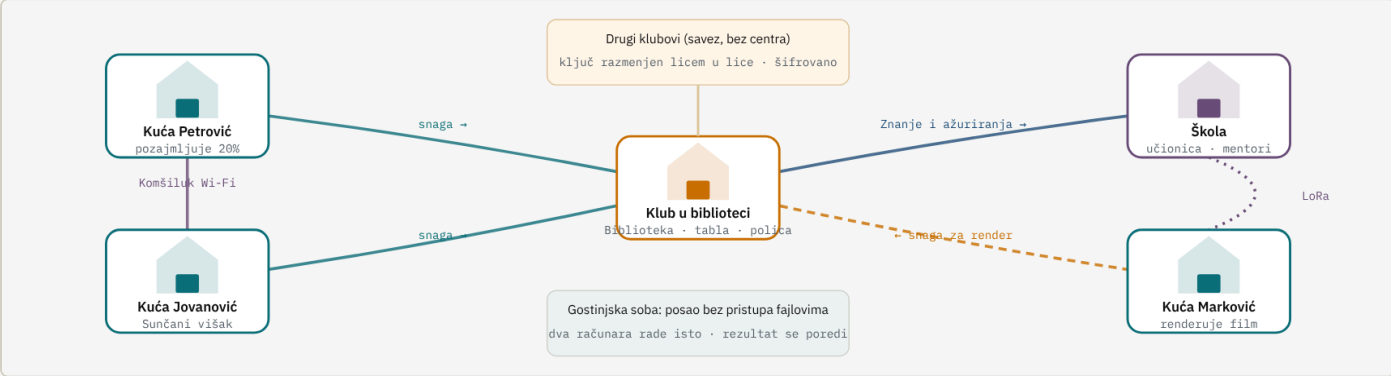
CHECKING

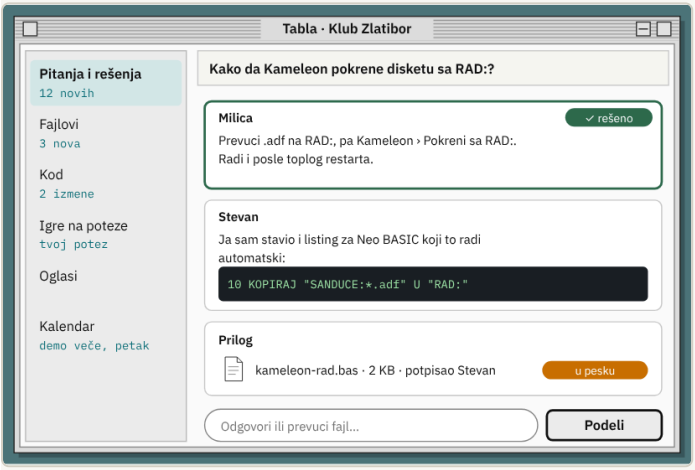
The Treasury checks duration and loudness (all themes equally loud) and that an error is clearly distinguishable from success. Themes may not contain ads or speech that sells something.

PART 2 · SOFTWARE

2.2e Co-op and the Board

The spirit of the old computer clubs and free software, built into the system: users share knowledge, results and part of their computer's power, voluntarily and without intermediaries. It all rests on the same foundation as the rest of Neo, namely Keys (permissions) and closed rooms, so sharing never opens your files.







Lending power

A computer that helps while it's idle

WHAT

An idle Neo lends part of its power to a friend, a club or science: rendering a film, translating a book, building a program, training a local model or a science project of your choice.

WHEN

Only when it is plugged in, when you aren't using it and, if you want, only from the Sun station's surplus. You set the limit for power, hours and temperature.

REWARD

Not money but thanks: whoever helped gets priority when they need the club's power.

Guest room

Someone else's job, your data untouched

SANDBOX

The borrowed job runs in a closed room on Fortress, without access to files, the camera, microphone or the rest of the network. It gets only the input, it returns only the result.

CHECKING

The same piece of work is done by two computers, and the results are compared. A wrong or fake result is seen at once.

Shared results

Done once, available to all

WHAT IS SHARED

Shader caches for games, OCR and translations of public domain books, map tiles, Neo packages and updates to the Knowledge cartridge, directly between users.

TRUST

Programs are built reproducibly, so anyone can check that the same source code gives the same file. A result is valid when the fingerprint matches from several independent sources.

Code

See how it works

SOURCE

Every Neo app has a “See how it works” button that opens its source code in Neo Kit. A change can be tried, saved as your own branch and shared with the club.

LISTINGS

Neo BASIC programs are shared like listings from old magazines: one file, runs right away, and you see right away how it was made.

Klub

Cyber computer club

WHERE SHE LIVES

On one Neo at someone's home, in a library, school or cultural centre. Clubs connect with each other without a central server, like a federation.

WHAT IT HAS

A shelf of programs, games and levels, its own Library, shared power, a message board in the spirit of a BBS, a calendar of meetups and demo competitions in Arcade.

PEOPLE

Mentors help newcomers through Help a Friend. The club chooses its own rules and moderators.

Neighbourhood

Even without the internet

NETWORK

Neo computers in the neighbourhood connect directly over Wi-Fi, and at a distance over LoRa radio for messages and small files, for villages and schools without good internet.

CARTRIDGE

Big things, like the whole Library, travel on a cartridge from hand to hand, signed so it's known they weren't changed.

Board

A BBS built into the system: sharing like network play in every game

EVERYWHERE

Just as every game in the Treasury has network play, every app has “Share with the club”: a document, a picture, a game level, a Neo BASIC listing, settings or an Arcade core. The file goes to the club Board in one move, with a description and licence.

SECTIONS

Questions and solutions (the answer that helped gets a “solved” badge, so Dijana sniffs it out the next time someone asks the same thing), Files, Code with change review and branches, turn-based Games like the BBS doors of old, and Ads for trading equipment.

MAIL ACROSS THE NEIGHBOURHOOD

Like the old BBS networks: messages and files travel from club to club when the computers meet, over the internet, the Neighbourhood network, LoRa radio or a cartridge. It works even where there is no internet.

LOOK

A modern view in Neo Silver, or a retro mode with text and ANSI colours, with modem sound if you like. Same content, you choose how it looks.

SAFETY

Every file is signed with the author's key and verified before opening. A program from the Board runs in a sandbox until two members sign it. The Board shows no ads and doesn't track who reads what.

Security

Share, and stay protected

MEMBERSHIP

You join by invitation, and keys are exchanged face to face via a QR code, like a handshake. Trust is built through people you already know.

SOMEONE ELSE'S CODE

A program from the club runs in a sandbox until two members review and sign it. Only then can it ask for wider permissions, and even then only with your consent.

CONNECTION

Everything is end-to-end encrypted. Lock's Personal and Work labels apply here too, so the system asks before anything labelled leaves the computer.

KIDS

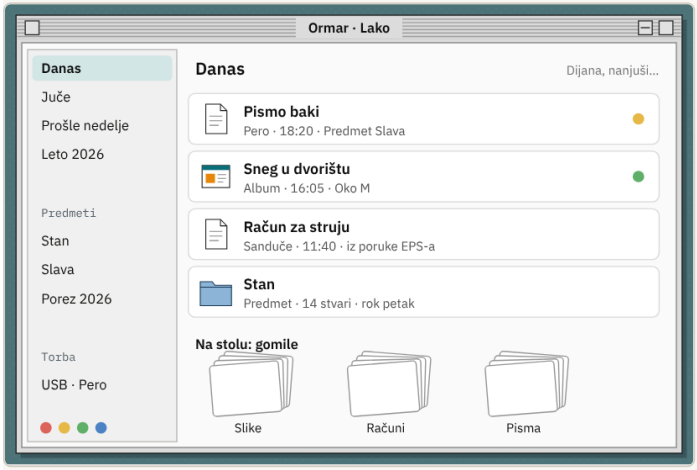
A child profile joins only clubs the parent approves, without power lending and without messages from strangers.

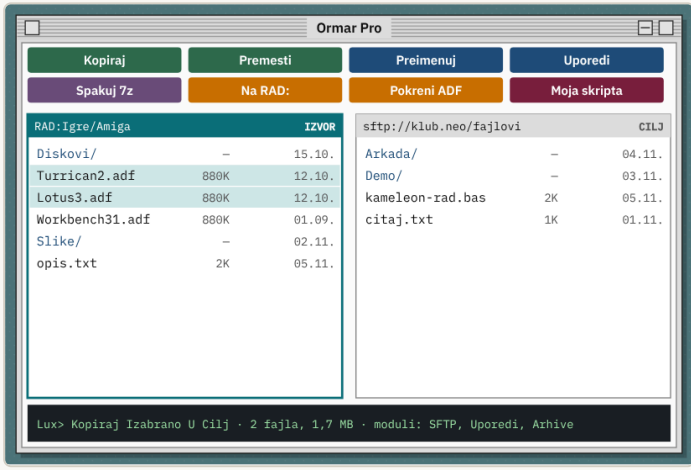
FREEDOM

Whoever leaves the club takes their data and programs with them. No account with us, no tracking and no ads.

2.2f Files without hassle and Cabinet Pro

Two faces of the same Cabinet. In Easy mode there are no files, extensions or saving, only documents, as on the Lisa: you work, and Neo remembers. In Pro mode Cabinet becomes a custom tool in the spirit of the last Directory Opus on the Amiga, with listers, button banks and modules.





No saving

No “Save” and no file names

REMEMBERING

Every change is remembered immediately, in RAD: and then on the SSD, and versions are made automatically. The “Save” button doesn't exist, as on the Lisa.

NAME

A document gets its name from the first line or title, and a photo from the place and date. You rename it when you want, you never have to.

NOVO

A new document is torn off a Template, like a sheet from a pad. You don't choose a program, you choose a kind of paper.

One place

A file lives in only one place

LINKS

“Also put in...” and “Send” make a link, not a copy. A change is seen everywhere, so there is no “contract_final_2_real”.

DUPLICATES

When a file arrives that you already have, Neo says “You already have this in Work” and doesn't make a second copy.

BAG

An external disk and a cartridge are a Bag: what you put in it, you carry with you, and Neo remembers what went where.

Views

By time, by person, by job

TIME

Today, Yesterday, Last week, Summer 2026. People remember when they did something better than where they put it.

PERSON

Everything from Grandma or from the accountant in one place: messages, files and pictures.

LIVE FOLDERS

A folder that is actually a question (“bills this year”) and fills itself.

On the desk

Piles, colours and Cases

PILES

Files on the desk stack into piles by type or day, like papers. Swiping a finger flips through a pile.

COLOURS

Seven colour labels with names you choose yourself, as on the old Mac.

CASE

A project folder that holds files, messages, notes, deadlines and people together: “Flat”, “Slava (family feast)”, “Taxes 2026”. Close the Case and everything is put away.

Simple

Little things that take away worry

EXTENSIONS

Hidden in Easy mode. A file is recognized by its icon and preview, and dangerous files never disguise themselves as documents.

SPRING

A folder opens when you hold a dragged file over it, so you go deeper without letting go.

NOTHING GETS LOST

Moving and deleting can be undone, the Bin keeps things for 30 days, and Dijana sniffs out where a file went.

Ormar Pro

In the spirit of Directory Opus on the Amiga

LISTERS

As many listers as you want, each with its own view, and any two become source and destination.

BUTTON BANKS

Banks you build yourself: a button runs a command, a Lux script or a series of steps, with left and right click.

FILE TYPES

Every type (picture, floppy, archive, code) has its own actions for double-click, drag and menu.

Modules

A Cabinet that grows

BUILT IN

FTP and SFTP as an ordinary lister, comparing and syncing two folders, batch renaming with a preview before running, finding duplicates, viewers for all formats and working with archives as with folders.

LUX

Every action has a Lux command, like ARexx on the Amiga, so Cabinet can be controlled from other programs and by voice.

SHARING

Banks, scripts and modules are shared via the club Board, signed and sandboxed until reviewed.

Two faces

Easy and Pro on the same files

SWITCHING

One switch, without losing settings. In Pro mode you see the real names, paths (Foundation is System:), extensions and permissions.

SAME DATA

In Pro mode a Case is an ordinary folder with labels, and a pile is a view, so nothing is ever locked into a format only Easy understands.

Lisa and beyond

More ideas from that school

CARD FILE

Contacts, notes and recipes in one system card file that every app reads, as on the Apple Newton of the nineties.

SEND AS A PACKAGE

“Pack for mail” shrinks pictures and compresses files to the mail limit by itself.

WHERE IT CAME FROM

Every file remembers where it came from (message, website, camera) and shows it with one click.

PART 2 · SOFTWARE

2.2g Neo Workshop: the Pro pack

The best free programs for graphics, 3D, layout, photography, sound and video, arranged as one professional pack. The programs stay free under the GPL licence and anyone can download them for free. The one-time purchase pays for convenience: Neo versions accelerated on our hardware, a consistent look, import of other formats, lessons and support. A large part of the money goes back to the authors.

Neo Workshop	For what	Engine	What Neo adds	Honestly: the weak spot
Sculptor	3D modelling, animation, rendering	Blender	Rendering on VSA-6000 via Vulkan, denoising on Neuron	Nothing significant, it's already top class
Brush	Painting and image editing	Krita, GIMP	Brushes on Newton, pen pressure, the same layout as Quill	Import of layered files from other packages isn't perfect
Vector	Logos, illustration	Inkscape	Fast display on Ray, PDF and EPS import	Complex foreign files sometimes lose effects
Typeset	Layout of books and magazines	Scribus	Cyrillic and Latin layout, ICC profiles of local print shops	Exchange with agencies that work in other packages
Darkroom	RAW photos	Darktable	Profiles for the Eye cameras, acceleration on Neuron	CMYK printing requires Typeset or Brush
Montage Pro	Video and effects	Kdenlive, Natron	Hardware decoding, effects on VSA-6000	Advanced effects and motion tracking are weaker
Director	AI film and images from a 3D scene	ArtCraft	Local models on Neuron, the scene on VSA-6000, a provenance label on every frame. Cloud models only with explicit consent and your own account	The best models are in the cloud and are paid separately. The engine's licence needs to be checked
Studio	Sound recording and mixing	Ardour	Orpheus under 5 ms, Neo microphone inputs	Fewer ready-made plugins

Price

Once and forever

ESTIMATE

€99 to €149 for the whole pack, with updates within the major version. Free for schools and libraries.

WITHOUT BUYING

The same programs without the Neo additions can be downloaded for free from the Treasury, because they are free software. The pack locks nothing in.

Fund

Money also goes to the authors

SPLIT

Proposal: half to the projects whose engines are used (the development funds of Blender, Krita and the others), a third to the team making the Neo versions, and the rest to local lessons and translation.

PUBLIC

An annual report on how much went to whom.

Code

All changes go back upstream

RULE

Every acceleration for Vulkan, Neuron or Orpheus is sent to the project, so it benefits everyone, on other computers too.

FORMATS

Work on importing other formats is the most useful gift to the community, because that is the main obstacle to switching.

PART 2 · SOFTWARE

2.2h Runs almost everything

The user doesn't choose an emulator, a compatibility layer or a virtual machine. Double-click a program or installer, and Neo itself chooses the room where it runs best, just like Arcade for games. Each guest lives in its own Guest room and sees only what you give it a Key for.

What you run	The room Neo chooses	How it works	Honest
Windows programs and games	Bridge + Wine/Proton	x86 code is translated to ARM on the fly, DirectX goes via Vulkan to VSA-6000	Most programs and games work, and the “Does it work?” database tells you in advance
Windows that doesn't work that way	Full Windows in a box	On Neo 2000 and 4000 with the Satellite x86 module at full speed, otherwise in emulation	The user buys the Windows licence
Linux programs	Apps domain	deb, Flatpak and AppImage, the windows behave like Neo windows	Almost everything
Android apps	Android room	Open Android in a separate room, stores of your choice, microG instead of Google services if you want	Apps that require Google device verification (some banks) don't work
DOS, Amiga, C64, consoles	Chameleon and Arcade	FPGA core if there is one, otherwise an emulator	The user brings the games and ROMs
Web apps	Horizon	Installed like ordinary programs, with an icon on the shelf	Everything that works in a modern browser
Games with kernel-level anti-cheat	None	That anti-cheat demands full control over the system, which clashes with Fortress	Doesn't work, unless the publisher allows running under Proton
macOS and iOS programs	None	The licence ties them to Apple hardware	Doesn't work
Streaming films at the highest resolution	Horizon	Requires certified content protection in hardware (Safe could do it, with a contract)	Lower resolution until certification

One click

The user doesn't choose the room

ORDER

Neo checks the “Does it work?” database, then tries Bridge and Wine, then the Android room or a full box. If nothing works, it says why, in plain language.

PRO

In Pro mode the room is chosen by hand, with settings like in the Proton tools.

Does it work?

A shared database of experiences

HOW

After the first run Neo asks “How did it go?”, and the rating and settings go into the database on the Board, like the shared compatibility boards for games on Linux.

PRIVACY

Only the program name, version and rating are sent, and only if you agree.

Security

The guest doesn't see the house

ROOMS

Each guest has its own part of the disk and sees your folders only when you give it a Key. A virus in the Windows room stays in the room, and Chronos takes it back to yesterday.

RULES

Lock labels apply to guests too: Personal and Work files don't enter the room without asking.

2.2i The best of the Unix world

We already have search, Chronos (going back in time), Flip Through, the app as a single file and Keykeeper. These are the next good ideas from the Unix world, under our own names and adapted to Easy and Pro mode.

macOS X

Looking and acting without opening

SPACE BAR

Select a file and press the space bar: a quick preview of a picture, PDF, film, sound or spreadsheet, without opening a program.

SERVICES

The Services menu offers actions on the selected text or picture in every app: translate, read aloud, send to Quill, make a task, sniff out.

RECIPES

A series of steps without programming (“shrink the pictures and send them to Grandma”), assembled by dragging blocks and run through Lux.

macOS X

Little things for the hands

TITLE ICON

The small document icon in the window title can be dragged into a message, onto the Board or into another folder, as if you were carrying the whole document.

HOT CORNERS

Mouse into a corner: Flip Through, the desk, Nap or the screen saver, as you choose.

DICTIONARY

A long press on a word opens the dictionary and Wikipedia, without the internet.

macOS X

Several Neo devices as one

CONTINUE

Start a letter on the Neo 500, continue on the Neo 2000 with one click. Clippings work on both.

HAND TO A NEIGHBOUR

A file is sent to a nearby Neo or phone through the Neighbourhood, without a cable and without the internet.

ONE MOUSE

The same mouse and keyboard move from one Neo to another across the screen edge, and a tablet becomes a side screen.

MOVING HOUSE

A wizard transfers files, mail, pictures and settings from the old computer, from Windows and Mac too.

DISK MODE

A Neo connected by cable becomes a disk for another computer, for rescuing data when the system won't start.

Darwin

Devices without setup

NEIGHBOURHOOD

Printers, scanners, speakers and shared folders on the network show up by themselves, as with Bonjour. No hunting for IP addresses.

PRINTING

Driverless printing over the open IPP standard, so it works with almost every new printer.

Darwin

A smart file system

CLONES

A copy of a file is instant and takes no space until it changes. Copying a 20 GB film takes a fraction of a second.

METADATA

Labels, colours, origin and notes travel with the file, on the Bag and through the Board.

DUAL PACKAGES

One neopak carries code for ARM and for Satellite x86, so the system picks the faster one.

Darwin

Order under the hood

HELPERS

One manager for all services, which start only when needed, like launchd.

MAGNIFIER

In Pro mode, live tracing of what the system and programs are doing: files, network, power. No restart and no special tools.

LOG

One system log, with personal data hidden by default.

Linux

Habits that save time

MOUSE

Middle click pastes the selected text, and Alt plus drag moves a window wherever you grab it. Switchable in Pro mode.

SHELL

Every command has short help with examples, like tldr, before the long manual.

SCREEN

Night light and power profiles (Quiet, Balanced, Turbo) from the menu.

Linux

Security without questions

PORTALS

A program doesn't see the disk. When it asks for a file, a system dialog opens, and the program gets only the file you choose. It's a Key you don't need to understand.

FIRMWARE

The firmware of all devices, external ones too, is updated through Renewal with a signature, like fwupd on Linux.

Linux

Neo in your pocket

LIVE SYSTEM

The whole Neo OS on a cartridge or USB disk, starts on another computer without installation, with your Workdesk.

TRY IT

A school or shop can let people try Neo on their own computer before buying.

2.3 Apps in the box

Need	Application	What it does	Engine
Reading	**Leaf**	PDF, EPUB, comics (CBZ), DjVu. Night mode, notes in the margins, reading aloud via Neuron.	Poppler, Foliate
Files	**Cabinet**	Two panels, buttons the user makes, batch renaming, preview without opening, archives, FTP and SFTP as ordinary folders.	libarchive, GVfs
Pictures	**Album**	Fast browsing of thousands of pictures, faces and places recognized locally, easy touch-ups, comparing two pictures.	libvips, darktable
Sound	**Wave**	Waveform and multitrack mode, noise removal, mastering, recording from the Orpheus input.	Audacity engine, LV2
Music	**Turntable**	Library, radio, podcasts, Cyrillic shown correctly in titles.	GStreamer
Video	**Cinema**	Plays every format, subtitles are found and translated locally.	mpv
Planner	**Planner**	Notes, calendar, contacts and reminders in one place.	CalDAV, CardDAV
Maps	**Atlas**	Maps that work without the internet, routes on foot, by bike and by car.	OpenStreetMap
Chat	**Chat**	Encrypted messages and video calls, without a phone number.	Matrix, Jitsi
Passwords	**Keykeeper**	Passwords and keys are kept in the Safe chip, never in the cloud without your consent.	KeePassXC format
Scanner	**Scanner**	Scanning and text recognition, Serbian Latin and Cyrillic.	Tesseract
Translation	**Interpreter**	Translating text and pages without sending data to the internet.	Bergamot, Neuron
Phone	**Link**	Files, messages and notifications between phone and computer.	KDE Connect
Retro	**Arcade**	Library of games and cores, cartridges and floppies in one place.	Chameleon, RetroArch cores
For the curious	**Shell and Neo BASIC**	A terminal that explains every command, and a language for first programs.	—

2.4 Good old ideas

Idea	From	In Neo OS
HyperCard	Apple, 1987	Neo Cards: apps made of cards and buttons without programming, for recipes, collections and school projects.
Services from the menu	NeXTSTEP	Select text or a picture in any program and choose what other programs can do with it: translate, send, convert.
Smart folders	BeOS	Files carry attributes like a database, and a folder can be a permanent search (“all invoices from 2026”).
Screens that slide down	Amiga	Pull the desktop down and underneath you see a retro core or the Windows layer from Panorama.
Window shade and spring-loaded folders	Mac OS 8/9	Double-click the title to roll the window up into a bar. A folder opens by itself when you hold a dragged file over it.
RAD: disk	Amiga	A working disk in RAM that survives a restart. The contents remain after updates and program crashes, and back then you could even boot from it.
Drag to save	RISC OS	The icon from the save window is dragged into a folder. No wandering through a dialog.
Look underneath	Smalltalk	Right-clicking any element shows which Lux command it runs, so every action teaches you how to automate it.
Command palette	modern editors	One shortcut opens a field where you type the name of an action, program or file, and it works in every app.
Focus mode	modern phones	One switch silences notifications and hides everything except the work you are doing.
Accessibility from day one	modern OSes	Screen reader, magnification, captions for every sound and voice control, turned on at first power-on if needed.

PART 2 · SOFTWARE

2.5 The Neo OS concept (in detail)

> Source document from 3 November. Adopted later: the Fortress microkernel underneath (Linux only as a driver domain), OS in ROM, Renewal instead of separate drivers, Guardian for backups, HAM modes dropped.

2026-11-03 ·

Short answer

Yes, such an OS could cover about 90% of modern needs, but only on one condition: that underneath its "Amiga/Haiku" personality it uses the Linux kernel. Every system that wrote its own kernel (AROS, MorphOS, Haiku) got stuck on drivers, not on the idea.

The advantage of this platform is that the hardware is fixed: we write the drivers for the Neo-Amiga chipset once and only for ourselves, and Linux brings everything else (USB, Wi-Fi, printers, Bluetooth). On top sits a completely custom, fast and consistent user system.

What remains out of reach or risky, and should be said up front:

- Games with kernel-level anti-cheat (Valorant, some competitive titles) won't work even through the Proton layer.
- DRM streaming in the browser (Netflix, Disney+ at full resolution) depends on a Widevine licence, which Google rarely grants to new platforms. The workaround is the Android layer.
- Professional closed software (Adobe, AutoCAD, some Microsoft Office workflows) runs only through the Windows layer, with varying success.

The names "Neo OS", "Exec-N" and others in the document are working names, aligned with the Neo-Amiga chipset from the neighbouring topic.

Philosophy: the best of six systems

Neo OS takes the speed and simplicity of the Amiga, the consistency of Haiku, and the hardware support and security of Linux. Each system gives what it is strongest at, and its weaknesses stay outside.

Sistem	What we take	What we leave
MorphOS	The feeling of instant response, the Ambient desktop, the Reggae multimedia framework, the self-arranging MUI-style GUI, polished apps (Wayfarer, Iris)	Closed source, PowerPC dependency, weak memory protection
AROS x64	The open Amiga API, portability, Zune (MUI clone), datatypes, assigns (logical drives like `Documents:`), ARexx ports in every app	Few drivers, single-processor legacy, shared address space
Linux	Drivers, security, Mesa and Vulkan, containers, Wine/Proton, Android via Waydroid	Desktop fragmentation, X11 legacy, packaging chaos, inconsistent look
Haiku	One consistent API (BeAPI), a thread per window, attributes and queries on files (BFS), packages that are "activated" without installation, a system that looks like a whole	Limited drivers, weak support for GPU acceleration

Two more role models come from the 8-bit and FPGA world:

Sistem	What we take	What we leave

SymbOS (Z80, Amstrad CPC, MSX, PCW)	Proof that pre-emptive multitasking, windows and networking fit into a few hundred KB: a message-passing microkernel, a memory bank manager, tiny apps. For Neo OS this becomes the "budget" for every system app and the yardstick for whether something is bloated. SymbOS itself runs natively on a Z80 core in the FPGA, and its windows appear as ordinary Neo windows	The limits of 8-bit hardware
MEGA65 OS (hypervisor and freezer)	A thin hypervisor under everything, a setup menu at boot (like a BIOS, but understandable), instant switching of machine modes and a freezer that freezes any program and saves it to disk. For us Chronos does this, for native programs and FPGA cores alike	Text interface and being tied to one machine

By "MEGA-OS" I understood the system of the MEGA65 computer (C64-Wiki); there is also a RealitySoft product of the same name, so if you meant that one, say so.

From this follow four design rules:

1. **Responsiveness before throughput.** Input, sound and window drawing have guaranteed priority; nothing in the background may stop the mouse.
2. **One system, one API, one look.** No choice between five desktops and three toolkits; third-party tools get the same look through compatibility layers.
3. **Every app is scriptable.** Like ARexx on the Amiga: every program exposes commands that scripts, other programs and (optionally) an AI assistant can use.
4. **Small and understandable.** The goal is for the base system to fit in a few hundred MB and boot in a couple of seconds.

Architecture

Neo OS is a Linux kernel with a completely custom user system, like Android and SteamOS, only with an Amiga and Haiku soul instead of Java or KDE. The user never sees a "Linux desktop"; they see Neo OS.

The coloured layer is what makes Neo OS different; everything below it is adopted and tuned, everything above adapts to it.

- **Exec-N** is the successor to the Amiga exec.library: message passing over shared memory (zero-copy, on memfd and io_uring), with real-time priority for input, sound and drawing. That gives that MorphOS feeling that the system is never late, but with memory protection and multiple cores.
- **Neo Kit** combines the Haiku BeAPI (a thread per window, a clean C++ API) with the Zune/MUI approach where the GUI arranges and scales itself. The compositor runs directly on the Ray and Panorama chips, with hardware layers for windows.
- **File system**: Btrfs with extended attributes and an index, so Haiku-style queries ("all PDFs from Marko from 2026") work across the whole disk. Amiga assigns (`Documents:`, `Games:`) are logical names on top of that.
- **Packages**: an immutable system image with A/B updates (like SteamOS) and Haiku-style packages that are activated without unpacking. Linux apps come as Flatpak.
- **Optimization for this ISA**: the whole system is compiled for exactly this processor (SVE2 or RVV vector extensions), memory is shared between the CPU and chipset so picture and sound go without copying, and Chronos enables snapshot and resume of any program.

Wayland as the display language. The Neo compositor speaks the Wayland protocol natively, so all Linux apps (GTK, Qt, SDL, Electron) run without an X server and get Neo window frames. Neo Kit uses the same protocol plus a few of its own extensions:

- `neo-menubar`: a global menu bar at the top of the screen, for native and foreign apps.
- `neo-layer`: a window gets its own hardware layer on Panorama (video, games, retro screens without copying).
- `neo-snapshot`: the compositor asks Chronos to freeze or restore a window together with its program.

Old X11 programs go through Xwayland, Wine has its own Wayland driver, and Waydroid already runs on Wayland. So every compatibility layer ends up as an ordinary Wayland window, which is the basis for the shared look below.

System structure: every directory has one purpose

Modelled on AmigaOS and BeOS/Haiku, the user sees a few clearly named areas, and the Linux tree (`/usr`, `/etc`, `/var`) stays hidden underneath. Every area also has an Amiga-style logical name (assign), so a script can say `Documents:Offers/` regardless of where it physically is.

Area	Logical name	Purpose	Who may change it
Sistem	`System:`	Immutable image: libraries, drivers (`System:Devices`), commands (`System:Commands`), startup scripts (`System:Scripts`), default settings	System updates only
Apps	`Apps:`	Every app is one folder or package; deleting the folder deletes the app without leftovers	Installer and user
User	`Home:`	`Documents:`, `Pictures:`, `Music:`, `Games:`, `Blocks:` (templates, see the Lisa section)	User
Settings	`Settings:`	The settings of every app as a readable text file, one folder per app, like Haiku `config/settings`	User and apps
Boxes	`Boxes:`	Linux, Android, Wine prefixes, ReactOS disk, Amiga `DH0:`, QL and Z80 disks	Each box only its own folder
Temporary	`RAM:`	A RAM disk as on the Amiga, emptied at shutdown	Everyone

The real names on disk are English (`System`, `Apps`, `Home`), and the displayed names are localized, as in macOS. So scripts and documentation work in every language, and the user sees their own language.

Error messages that mean something

Every error message answers three questions: what happened, why, and what the user can do now. That is the spirit of AmigaOS requesters like "Please insert volume Work in any drive" and BeOS's care for language, without the Guru Meditation code nobody understood.

Instead of	Neo OS says
Error 0x80070005	Quill can't save "Offer.odt" because the folder Documents:Archive is write-protected. [Save elsewhere] [Unlock folder]
Unknown error	Pigeon can't connect to the server mail.example.com: the server hasn't answered for 30 seconds. Check the network or try later; the messages stay in the Outbox.
Segmentation fault	Abacus stopped because of an error in the program. Chronos saved the state from 12 seconds ago. [Restore state] [Send report]

Rules for developers and the system:

- Every message has a short code (for example `NEO-FS-017`) that leads to a local help page, and a "Details" button for the technical record.
- Messages without a suggested next step are forbidden; Neo Kit won't show them without a "what next" field.
- A program crash never deletes work: the Chronos snapshot and autosave (see the Lisa section) restore the state.
- When Claude is on, the "Explain" button gives an explanation in plain language for exactly that record, but the message must be clear without it.

- Guru Meditation remains only as a tribute, on the kernel crash screen, but with clear text and a QR code to the report.

Compatibility layers and their overhead

The lowest overhead comes not from emulation but from reimplementing the API: the program is compiled for our processor and thinks it is on its own system. Emulation remains only for programs whose source code doesn't exist or which are for another processor.

Software source	Method	Overhead (rough estimate)	Reality
Linux ARM64 / RISC-V	Native, same kernel and libraries	~0%	Trivial
Android apps	Waydroid container, ARM64 native	~5%	High; solves banking and streaming apps
Haiku apps	BeAPI reimplementation on top of Neo Kit, recompiling; Haiku code is MIT-licensed so much of it can be reused	~0%	High
AROS apps	Amiga API libraries (exec, intuition, Zune) as a thin layer on top of Exec-N, recompiling	~0-5%	High; AROS already exists for ARM
MorphOS apps	PowerPC JIT + reimplementation of the MorphOS ABI (clean-room, since MorphOS is closed)	2-4x slower, but the programs are light	Medium to low; the biggest job of all the Amiga layers
Classic Amiga (68k)	68030 + AGA core in the FPGA fabric, or UAE JIT in a window	~0% (FPGA, cycle-exact)	High
Sinclair QL and 8-bit computers	FPGA cores (68008, Z80)	~0%	High; the QL Psion suite literally runs
Linux x86-64	Native on the x86 coprocessor; fallback dynamic translation (FEX or box64 style) on ARM	~0% on the coprocessor; 20-50% through translation	High; practically solved
Windows x86/x64	Wine + the same translation + DXVK/VKD3D for graphics	~5-10% on the x86 coprocessor; 20-50% through translation on ARM	Medium to high; see the Steam section

The key idea for this ISA: the x86 translator uses hardware assists proposed in the CPU topic (for example a TSO memory model as in Apple M chips), which pushes the loss to the low end of the estimate. The FPGA fabric is loaded on demand through a system service (working name Chameleon): 68030 today, Z80 tomorrow, and when there is no retro work, a video codec accelerator.

x86-64 coprocessor: hardware Windows and x86 compatibility

As agreed in the CPU topic, besides the main ARM processor and the FPGA block the board also has a separate Intel/AMD-compatible chip. Neo OS doesn't use it as a second main processor, but as a "satellite" for x86 software:

- The x86 chip runs a small, invisible Linux kernel (working name Satellite) that runs only Wine/Proton processes, x86 Linux programs and the ReactOS box.
- The windows and sound of those programs go back to the Neo compositor through shared memory, so the user sees no difference. Vulkan calls are forwarded to Mosaic; this works well only if the link between the chips is coherent and fast (PCIe 5 or CXL class).
- The system scheduler (working name Dispatcher) decides by itself where a program runs: natively on the x86 chip, or on ARM through translation when the x86 chip is busy or turned off to save power.

This is the answer to the Atari Jaguar worry: the Jaguar and the PlayStation 3 didn't fail because of multiple chips, but because programmers had to use them by hand. Here an app never knows which chip it is running on; the OS carries all the complexity. The cost remains: higher power use, a more expensive board and one more set of drivers for the Satellite link.

ReactOS box: a minimal Windows sandbox

Besides Bridge (Wine) the system also offers the ReactOS box: ReactOS, an open clone of the Windows NT kernel, in an isolated virtual machine with one window and a shared folder. Wine remains the main route because it is faster; the ReactOS box is for what Wine can't do.

- **When it's used**: old installers, programs that need a real NT kernel or their own drivers, USB dongle protection, old devices with Windows drivers, and running a suspicious .exe file without risk to the system (the box is simply thrown away afterwards, or restored with a Chronos snapshot).
- **How it works**: the ReactOS box runs directly on the x86 coprocessor, with hardware virtualization (VT-x or AMD-V), so it is close to native speed. Without the coprocessor (cheaper model) it falls back to full-system emulation, 5-10x slower (rough estimate).
- **FPGA shortcut**: for DOS and Windows 9x a 486 core can be loaded into the fabric (like ao486 in the MiSTer project), so that generation runs without software emulation.
- **Reality**: ReactOS is still alpha software and doesn't run everything Windows XP runs. So the box is marked "experimental", and for serious cases the user can still put their own Windows licence into the same box.

Visual look: Neo Platinum

The look is based on the spirit of Mac OS 8 and 9: a light grey surface, bevelled edges, stripes in the title of the active window, a global menu bar at the top and a control strip in the bottom left corner. Neo adds its own: a blue accent, origin badges on frames (Wine, FPGA), a Claude switch in the menu bar and Chronos in the control strip. A desktop mockup and control panel are in Neo OS look; the Claude switch in the menu bar there really turns the palette on and off.

One look for all sandboxes

The rule is that the user doesn't see where Neo OS ends and a sandbox begins: the same frames, menus, dialogs and fonts in every box. This is achieved with six shared mechanisms, not by reworking every program.

Element	How it is shared	Which boxes
Window frames and buttons	The compositor itself draws the frame for every window (server-side decorations)	Everyone
Menu bar at the top of the screen	Menus are exported to `neo-menubar`: Linux apps via DBus menus, Wine and ReactOS via an agent that reads the Win32 menu	Linux, Wine, ReactOS, Amiga API
File, print and colour dialogs	One Neo dialog behind xdg-desktop-portal; in Wine and ReactOS a replacement for `comdlg32` calls the same portal; Android gets it as a documents provider	All except FPGA cores
Control theme	The same theme exported in several formats: GTK, Qt, Windows `.msstyles` (Wine and ReactOS support them), Zune/MUI, Android overlay	All except FPGA cores
Fonts and scaling	One set of fonts and one scaling factor for all	Everyone
Clipboard, drag and drop, notifications	Wayland and the portal for ordinary boxes, a guest agent for the virtual machine	Everyone

Two boxes need special treatment. The ReactOS box is a virtual machine that knows nothing of Wayland, so a small agent (working name Clasp) runs in it and reports the position of every window; the compositor then

shows them individually, without the ReactOS desktop, like VirtualBox's seamless mode. FPGA cores (Amiga, QL, Z80) keep their authentic look inside the window, and only the frame, menu and snapshot come from Neo OS.

Games: native Steam and the Proton layer

Steam on this platform is achievable, but only in agreement with Valve. Valve has already started down that road: for the ARM device Steam Frame (announced in November 2025) it uses Proton together with the FEX translator for x86 games, so Neo OS could be a "SteamOS-compatible" platform instead of building everything itself (from memory, not checked for this document).

What it would look like:

- **Steam client**: the ARM64 version Valve is already developing, run as a Linux app with the Neo OS window look and Big Picture mode for the TV.
- **Proton layer (working name Bridge)**: Wine + DXVK (DirectX 9-11) + VKD3D-Proton (DirectX 12) + an x86 translator. It is all open source, the job is integration and tuning.
- **The Vulkan driver for Mosaic is the critical point.** All of Proton stands on Vulkan. Without a mature Mesa driver for Mosaic/Ray, no Windows game will run well, regardless of everything else.
- **Chronos snapshot**: suspend and resume of any game, like a save state in an emulator, and instant recovery after a crash.

What to realistically expect:

Type of game	Expectation
Linux and ARM64 native games	Full speed
GPU-bound Windows games	Close to native, if the Vulkan driver is good
CPU-bound Windows games (strategy, simulation)	Close to native on the x86 coprocessor; 20-40% slower through translation on a model without one
Games with EAC or BattlEye protection	Work only if the publisher enables them for Proton
Games with kernel anti-cheat	Don't work, and won't

A special trump card of this platform is retro games: FPGA cores for the Amiga, QL and Z80 computers give an accuracy that software emulators on a PC don't have, and this can also be offered as a Steam library of "classics".

Basic app suite

The suite follows the spirit of the Psion Xchange suite for the QL (Quill, Abacus, Archive, Easel): four small, fast apps with the same menus and the same way of working, plus a browser and mail at the MorphOS level. The goal is for every app to start in under half a second.

The "Quartet" office suite (working names)

Application	Role	Psion ancestor
Quill	Word processor with styles, footnotes and PDF export	Quill
Abacus	Spreadsheets with formulas and charts	Abacus
Archive	A simple database and forms, queries on file attributes	Archive
Canvas	Charts, diagrams and drawings	Easel

The native format is ODF, with DOCX/XLSX/PPTX import and export. Realistically: perfect fidelity with complex Word documents is a decade of work, so for heavy documents the system offers LibreOffice as a Linux app. The light suite covers 90% of home and small business needs, LibreOffice the rest.

Stage: presentations, the fifth member of the suite

- Slides are made from blocks from the other apps: a table from Abacus, a chart from Canvas and text from Quill stay live and update when the source changes.
- Transitions and animations go through Copernicus and Ray, so they are smooth even on a big screen, and a presentation on a second monitor runs via Panorama with speaker notes on the first.
- "Outline" mode: the whole presentation is written as a list of headings and sub-points, and the look is chosen at the end.
- PPTX and ODP import and export, export to PDF and to a standalone HTML package for the web.

Browser (working name Horizon)

- Based on WPE WebKit, the version of WebKit made precisely for embedding on Linux. The Linux base saves years of work here; MorphOS's Wayfarer and Haiku's WebPositive show how expensive a port is without it.
- Its own interface in Neo Kit, GPU rendering through the Ray compositor, hardware video decoding.
- The biggest ongoing cost is security patches: WebKit must be tracked every couple of weeks, forever.
- Weak points: Widevine DRM (Netflix, Spotify web) and sites that work "only in Chrome". For DRM streaming the solution is Android apps through Waydroid.

Mail (working name Pigeon), in the spirit of MorphOS Iris

- IMAP, SMTP and JMAP, OAuth2 login for Gmail and Outlook (requires app registration with Google and Microsoft).
- Every message is a file with attributes (from, subject, date, labels), as in Haiku, so "smart folders" are just saved file system queries and search is instant.
- Calendar and contacts via CalDAV/CardDAV, PGP signing built in.

Alongside come small system tools in the same style: a viewer for any format via datatypes, a player on the Reggae framework and a shell that understands both AmigaShell commands and bash.

Task-oriented work: the return of the Lisa concept

Yes, the Lisa concept can come back, without giving up the apps listed: they stay in the background as tools, and the user works with documents and tasks. The Lisa (1983) had "stationery pads": the user didn't launch a program, but tore a new sheet off the "Letter" or "Spreadsheet" pad, and the system opened the right tool by itself.

What it looks like in Neo OS:

1. **Pads instead of programs.** The `Blocks:` folder contains templates (Letter, Offer, Invoice, Presentation, BASIC game). A double-click tears off a new document and opens Quill, Abacus, Stage or another tool, without asking which program.
2. **No "Save".** Everything is saved automatically, with version history (Btrfs snapshots), as on the Lisa. The menu has "Revert to version from" instead of "Save as".
3. **Tasks as a workspace.** A task (for example "Offer for Marko") gathers documents, messages from Pigeon, tabs from Horizon and notes. Switching to another task puts away all the windows of the first, and switching back restores them exactly as they were, with Chronos.
4. **A document knows its tool and its links.** Through file attributes (Haiku style) a document remembers which task it belongs to, who it came from and where it was sent, so the search "everything around the offer for Marko" is a single query.

A lesson from history: Apple's OpenDoc (1990s) tried to abolish applications completely and embed components inside each other, and failed because of complexity. So Neo OS keeps ordinary applications, and the document and task are just a layer above them; whoever wants the classic way of working with programs still has it.

Neo BASIC and Neo Studio: programming for everyone

It makes sense, and it may be the biggest trump card of the whole platform: a computer you switch on and can program straight away, like Sinclair and Commodore BASIC, with the power of AMOS and the ease of the CanDo and Hollywood tools. One program has two views that switch with one click: text (Neo BASIC) and visual (Neo Studio, cards and blocks like CanDo and HyperCard).

- **Ready at once.** The BASIC window opens with a shortcut and works in direct mode: `PRINT 2+2` gives the answer without a project or compiling.
- **Commands for our hardware**, as AMOS had for the Amiga: sprites and layers (Panorama), Copper lists (Copernicus), sound (Orpheus), physics (Newton), snapshot (Chronos).
- **The visual part**: windows, buttons and cards are drawn with the mouse in Neo Kit style, and events are written as short BASIC blocks or assembled from ready-made blocks.
- **Controlling other apps** through Lux ports, like ARexx from AMOS: a program can open Abacus, fill in a table and send it with Pigeon.
- **Claude, if it is on**, explains an error in the program and suggests a fix, but BASIC works fully without it.

```
SCREEN 1, 640, 360
SPRITE 1, "ship.png", 320, 300
REPEAT
  IF KEY("left") THEN MOVE 1, -4, 0
  IF KEY("right") THEN MOVE 1, 4, 0
  FRAME
UNTIL KEY("esc")
```

The commands also exist in Serbian (`EKRAN`, `SPRAJT`, `PONAVLJAJ`); the same program is shown in the language the user chooses.

Export to other platforms

On Neo OS a program runs in a fast virtual machine with a JIT compiler. For export it is compiled to C on top of SDL3 and Vulkan, the route by which Hollywood and similar tools already reach several systems.

Target	How	Reality
Linux (x86-64, ARM64)	A single executable or AppImage	Easy
Windows and macOS	The same C code, Neo OS compiles it itself (macOS signing requires an Apple account)	Easy to medium
Android	APK via the Android NDK, signed with the user's key	Medium
iOS	An Xcode project; compiling and publishing need a Mac and an Apple Developer account	Medium, Apple's rules are the obstacle
Web	A WebAssembly package that runs in every browser	Easy

Other platforms have no Copernicus or Chronos, so the export replaces them with a software version or turns them off, and Neo Studio warns in advance which commands won't work.

Backups: Blu-ray archive and cloud

Blu-ray plus cloud is a good solution, but not ideal on its own: Blu-ray is excellent for a lasting archive of photos and documents, but too slow and too small for a copy of the whole system. So Neo OS follows the 3-2-1 rule: three copies, on two different media, one of which is off-site.

Layer	Medium	What it keeps	When
Instant	Btrfs snapshots on the same disk	The whole system and all document versions	Continuously, automatically
Local	External SSD or HDD	The whole system and data, incrementally (Btrfs send)	Every time the disk is connected
Archive	Blu-ray BDXL, preferably M-Disc (100 GB per disc)	Photos, documents, finished tasks	When 100 GB of new data has accumulated, or manually
Off-site	Cloud, encrypted on the computer before sending	What the user marks, plus system settings	Every night

Blu-ray in the system (working name Archivist)

- One "Archive" button: the system chooses what is new since the last disc, writes it, verifies every byte and prints a short table of contents for the sleeve.
- Every disc carries a catalogue of the whole series, so a search for "photos from Kotor 2024" says which disc they are on, even without inserting the disc.
- Error correction across several discs (a couple of extra discs for every ten), so a damaged disc doesn't mean lost data.
- Risk: in 2025 Sony stopped making recordable Blu-ray discs, and the other manufacturers are few (from memory, to be checked). So the same Archivist must also work with LTO tape or ordinary disks as an archive.

A cloud that cooperates with Apple and Google accounts

- The backup is encrypted on the Neo computer, and the key stays with the user (and on a printed recovery sheet). Behind that can be any S3-compatible service or Neo's own cloud.
- The most reliable path for photos from a phone is a small Neo app for iOS and Android that sends new pictures to the Neo computer or the Neo cloud by itself. It doesn't depend on anyone else's API.
- Google: Google Takeout can be scheduled for a regular export of the whole account. In 2025 Google narrowed the Photos API, so reading the whole library directly is no longer possible (from memory, to be checked).
- Apple: iCloud Photos has no public API for reading. There is only Apple's official request for a copy of your data and transfer to Google Photos, so for Apple we rely on the Neo app on the iPhone.

This requires two more agreements from the licences section: with Google and Apple to publish Neo apps in their stores, and with the cloud operator.

Exchange with iOS and Android devices

The phone should be an extension of the Neo computer, not a separate world. The basis is a small Neo Link app for iOS and Android, with open protocols other systems already use, and Android through the Waydroid layer on the computer itself.

Feature	Android	iOS	How
Sending files both ways	Yes	Yes	LocalSend protocol (open), drag onto the phone icon on the desktop

Automatic photo import	Yes	Yes, while the app is active in the background	Neo Link, with the cloud from the backups section
Shared clipboard	Yes	Partly, iOS limits background access	Neo Link
Phone notifications on the computer	Yes	Limited	Like KDE Connect
Replying to SMS and calls	Yes	No, Apple doesn't allow it	Neo Link
Phone as camera, microphone, remote	Yes	Yes	Neo Link
USB copy and import without a network	Yes (MTP, ADB)	Yes (libimobiledevice)	Built into the system
Quick Share and AirDrop	Possible, the protocol is partly open	Only if Apple opens AirDrop	Depends on the manufacturer

On the iOS side the limits are Apple's, not technical. The EU Digital Markets Act (DMA) is pushing Apple to open some of these functions to other manufacturers, so the iOS list may grow (from memory, follow the European Commission's decisions). Contacts and calendar aren't transferred separately: they go over the CalDAV and CardDAV standards that both iOS and Android support.

Integrated Claude AI as an option

Claude is built in as a system service with one switch in the menu bar, and is off by default. When it is off, no assistant process runs and nothing goes to the network.

Three levels, changed with one click:

Level	What it does
Off	The assistant doesn't exist in the system; no background process and no network traffic
On call	Opens with a shortcut or from the menu, sees only what the user explicitly gives it (selected text, a file, a window)
Contextual	Can read the active document and suggest actions, still with confirmation for every change

The most interesting part is the link with the Amiga heritage. Since every app exposes an ARexx-like port with commands, the system automatically turns those ports into tools for Claude (in MCP form). So the assistant can tell Abacus "sum column C and make a chart" or Pigeon "find all the bills from October", without a special integration in every app.

Privacy and control rules:

- Permissions per app, as for the camera and microphone.
- Every action of the assistant is written to a local log that the user can review and delete.
- Deleting, sending mail and purchases always require explicit confirmation.
- Claude runs in the cloud and requires an account or API key. For offline work (dictation, local search, translation) a small local model on the Neuron NPU is used, clearly marked as local, not as Claude.

A people's, self-managed model: an OS owned by a co-operative

Neo OS can be anarchist and people's-democratic in a practical sense: there is no owner who decides alone, the user is master of their own computer, and decisions are made by those who build and use the system. There are proven models for this: Debian's social contract and constitution, Haiku Inc. as a non-profit, KDE e.V. and the federated Mastodon.

On the computer: user sovereignty

- No mandatory account, no telemetry and no ads. Every outgoing connection is visible in the settings and can be turned off.
- Full access to the system, an unlocked boot and the right to install any other OS. The board schematics are public, for repairs.
- The app store is federated, like F-Droid repositories: anyone can open their own, and the system comes with a few trusted ones.
- Updates are also shared over a P2P network between users, so the system doesn't depend on one server.

In the project: self-management

1. **A co-operative, not a company.** The Neo co-operative has members (developers, users, clubs), and every member has one vote, no matter how much they invested. Revenue from hardware and membership goes into development.
2. **Open decisions.** Every bigger change goes as a public proposal (RFC), with a deadline for discussion and a vote. Technical decisions are made by the teams for their part of the system; bigger decisions, like licences and the direction of development, go to a general vote.
3. **Rotation, not a boss.** Team coordinators and the project representative are elected for one year, like the Debian project leader, with a limited number of terms.
4. **Local clubs.** User clubs, like the Amiga clubs of old, have their representatives in the co-operative's assembly and translate the system into their language.
5. **Right to fork for apps.** Apps are open source (GPL or MIT), so anyone can change them and develop them further. The OS itself is closed (see the licence model below), and the protection against abuse of power is the co-operative's obligation to publish the OS source code if the project shuts down or is sold (escrow clause).

Licence model: closed OS, open apps

This is the model of MorphOS and partly of macOS, and it is achievable, with one limit: parts taken under a copyleft licence must stay open.

Part	Status	Reason
The Linux kernel and driver changes in it	Open	GPL-2 requires it
Wine, WebKit, Mesa, the Proton layer	Open (changes to them)	LGPL and similar licences
Modified AROS files	Open	The APL requires publishing changes to those files
Exec-N, Neo Kit, compositor, Chameleon, Dispatcher, Chronos service	Closed	Our own code; Haiku (MIT) parts may be closed
Apps (Quartet, Stage, Horizon, Pigeon, Neo BASIC)	Open	Project decision
Neo Kit API and documentation	Public and free	So that others can write apps

The advantage of a closed OS is a simpler deal with Valve, Google and others and protection of the work on chipset drivers. The price is that the community can't fix the system itself, so the public API, open apps and the escrow clause carry part of the trust that open source would otherwise carry.

Where the ideal collides with reality

The agreements from the licences section (Valve, Google, Microsoft, codecs) require a legal entity that signs and is accountable, so the co-operative must exist as a registered organization. The closed software that comes from that (Widevine, Steam, ROMs) is kept in a separate, clearly marked repository that the user turns on

themselves, like Debian's `non-free`. So the core of the system stays free, and the user chooses how many compromises to accept.

Licences: who to get permission from

Real negotiations are needed with about fifteen companies and organizations; the rest of the system is open source that only requires respecting the licence. The list was compiled from general knowledge, so the rights holders and terms should be checked with a lawyer before any launch.

Rights holder	For what	Note
Valve	Steam client, Proton and the ARM64 version of Steam	A partner agreement; without it Steam remains only through translation
Google	Widevine DRM for the browser; Google Play and Google services in the Android box	Without certification the Android box runs without the Play Store (F-Droid, Aurora)
Microsoft	Windows licence for real Windows in the box; possibly Windows on ARM	Most realistic is for the user to bring their own licence
Epic Games (Easy Anti-Cheat), BattlEye	Anti-cheat support for Proton on a new processor	The game publisher still has to enable it
Netflix, Disney+, Spotify and the like	Device and app certification for HD streaming	Depends on the Widevine deal with Google
Cloanto (Amiga Forever)	Kickstart ROM and Workbench for the Amiga FPGA core	The AROS ROM is a free replacement, but less compatible
Hyperion Entertainment	AmigaOS 3.x and 4 if offered as an option	The rights have been the subject of disputes; check the current situation
MorphOS Team	Consent or a licence for the MorphOS ABI and libraries	Without an agreement, only a clean-room reimplementation
Amstrad (today Sky)	ROMs for the ZX Spectrum and Amstrad CPC	Amstrad has previously allowed distribution with emulators; check the terms
Rights holders for the Sinclair QL ROM and Psion Xchange	The original QL system and the Psion suite	The rights holder needs to be determined
The MEGA65 team (MEGA e.V.) and Cloanto	MEGA65 ROM and C64 ROM	The MEGA65 core is open, the ROMs have separate terms
Prodatron (author of SymbOS)	Shipping SymbOS with the system	A personal agreement with the author
Via LA, Access Advance, Dolby, DTS	Patents for H.264, AAC, HEVC; Dolby Atmos and Vision, DTS	AV1, Opus and VP9 are royalty-free
Anthropic	A commercial API agreement for integrated Claude	Billing per use or through the user's account
Bluetooth SIG, Wi-Fi Alliance, HDMI Licensing, USB-IF	Hardware certification	Not the OS, but without it a device can't go to market

No negotiations, only respecting the licence: Linux (GPL-2), Wine and ReactOS (LGPL and GPL), FEX and box64 (MIT), DXVK (zlib), VKD3D-Proton (LGPL), WebKit (LGPL and BSD), Mesa (MIT), Haiku (MIT), AROS (APL), Waydroid and LineageOS (Apache and GPL). For patents around Linux (for example exFAT) the solution is membership in the Open Invention Network.

Realistic estimate: effort, risks, order

The whole concept is achievable for a team of 60 to 100 people in 4 to 5 years, and the biggest single risk isn't the OS but the Vulkan driver for Mosaic. The numbers below are a rough estimate by analogy with similar projects, not a calculation.

Part	Effort (person-years)	Risk	Why
Drivers for the Neo-Amiga chipset, especially Vulkan for Mosaic	40-70	High	Without it there is no Proton, no browser acceleration and no modern games
Exec-N, Neo Kit, desktop and compositor	20-30	Medium	Our own code, but it builds on Haiku (MIT) and AROS code
Horizon browser	5-8 per year, permanently	Medium	WPE WebKit exists; the cost is patches and site compatibility
Quartet office suite	15-25	Medium	The light part is quick; fidelity to the DOCX format is expensive
Proton layer and x86 translation	5-10	Low to medium	Mostly integration of existing open source, with Valve's agreement
MorphOS PowerPC layer	5-10	Medium	Clean-room work on a closed system, a small audience
AROS and Haiku layers	3-6	Low	Open source, the same API style
Pigeon mail and Claude integration	4-6	Low	Clear scope, existing standards

Proposed order, so that every phase already gives a usable system:

1. **Foundation**: Linux kernel, chipset drivers, Exec-N and Neo Kit, desktop and shell. At the end of the phase every native Linux ARM64 program runs.
2. **Everyday work**: Horizon browser, Pigeon mail, Quartet, the Android layer. At the end of the phase the system covers office work, the internet and banking.
3. **Games**: the Vulkan driver to full maturity, the Proton layer, Steam in agreement with Valve, Chronos snapshots for games.
4. **Heritage**: FPGA cores for the Amiga, QL and Z80, the AROS and Haiku layers, and finally the MorphOS PowerPC layer.
5. **Assistant**: ARexx ports as MCP tools and the Claude switch. Technically it could come earlier, but it is worth most when all apps already have ports.

Wayland is adopted as the display standard from the first phase (see Architecture), because only that way do the system and all the sandboxes look like a whole.

PART 3

3.1 Path to a hardware standard

The goal is for Neo to become what the IBM PC and MSX were: a specification by which several factories build compatible computers. World standards are adopted, and our own is made only for what is new. Timelines are estimates.

Phase	When	Hardware	What becomes the standard	Who manufactures
0. Paper and emulator	year 1	A software emulator of the whole system and an FPGA development board	Neo Hardware Manual 0.9: Copernicus, blitter, Panorama and Orpheus registers, cartridge mechanics	Universities, the demoscene, enthusiasts
1. Reference board	years 1 to 2	An off-the-shelf ARM chip and FPGA on one board, cartridge slot, DE-9, Safe	Neo Base 1.0: the mandatory minimum every Neo must have	The first local run, a few thousand units for schools and clubs
2. The first Neo package	years 2 to 4	Our own chiplets on an older process (12 to 28 nm), a shared pilot run, then a real production run	Neo Full 1.0: VSA-6000, Ray, Courier and Chronos. A certificate and the “Neo compatible” logo	Two or three licensed board and case manufacturers
3. Standard	years 4 to 6	A modern process, the 500, 2000 and 4000 family, third-party AGP-X cards	Neo 2.0, backward compatible. The standard is run by the Neo consortium	National factories under licence, cards and cartridges from several companies

We adopt

World standards, unchanged

LINKS

PCIe, CXL, UCIe for chiplets, USB-C with power delivery, M.2 and NVMe, DisplayPort and HDMI, COM Express as the basis for the Satellite module.

SOFTWARE

Vulkan, UEFI and ACPI following the ARM SystemReady rules, so another OS can be booted on Neo too. zstd, LZ4 and EROFS for compression.

We make

Only what is new

CONNECTORS

The Neo cartridge (mechanics, NVMe and signature), AGP-X (PCIe ×16 + CXL + a channel for Panorama layers), the Satellite connector, the DE-9 joystick pinout.

FORMATS

Custom chip registers, the signed Chameleon core format, neopak, the Chronos snapshot and the Neo ROM layout.

Two levels

Neo Base and Neo Full

BASIS

ARM core, FPGA fabric with Chameleon, Copernicus, Panorama, Orpheus, Courier, Safe, Neo ROM, cartridge slot, privacy switches. Enough for Neo OS, Arcade and school.

FULL

Adds VSA-6000, Ray, Newton, Neuron and Chronos. Programs ask the system what is present, so they run on Base too, only slower.

Certificate

Neo test suite

CHECKING

An open set of tests: registers, retro core timing down to the clock cycle, cartridge, Renewal, Lock and switches that really cut the power.

LOGO

Whoever passes the tests and pays for the certificate may use the name “Neo”. Whoever doesn't pay may build a compatible computer under their own name.

Governance

Neo consortium

WHO

A non-profit foundation with members: manufacturers, universities, schools, user clubs and countries that adopt Neo.

HOW

Public change proposals, voting, a new version of the standard at most once every two years and ten years of backward compatibility.

Licences

Open, but protected

SCHEMATICS AND SPECIFICATION

Schematics under CERN-OHL-S, the specification under CC BY. Anyone may read, learn and build.

NAME

The “Neo” trademark and Dijana belong to the consortium, as with Arduino: open hardware, protected name.

Manufacturing

What is made at home

LOCAL

Boards, cases, keyboards, cartridges, assembly, testing, service and FPGA cores.

IMPORTED

Silicon is made in foreign chip foundries, because that is the only realistic route today. The chip design stays ours.

Independence

No single point of failure

PROCESSOR

ARM requires a licence, so the Neo standard is written above the instruction level. A RISC-V core is the licence-free fallback, and Neo Kit programs are simply recompiled.

FOUNDRIES

Chiplets on proven processes offered by several foundries, so changing foundry isn't the end of the project.

Precedents

How others succeeded

COMPUTERS

IBM PC: published documentation gave birth to the clone industry. MSX: one standard, dozens of manufacturers. Galaksija: a schematic in a magazine.

TODAY

Arduino and Raspberry Pi (open and cheap), Framework (repairable), RISC-V and OCP (a community-run standard).

PART 3

3.2 Prices and the path to production

Model	Price	What you get
NEO 500	700 – 900 €	A computer in a keyboard, Eye S, OS in ROM, battery and 5G optional.
NEO 2000	1.600 – 2.000 €	Desktop, Eye M, large FPGA. The Satellite x86 module can be added for €250 to 400.
NEO 4000	3.500 – 4.500 €	Tower, Eye Pro, built-in x86, HBM, Blu-ray. Workstation class.

- Estimate: about 2× the cost of parts, at 100,000 or more units a year. Not checked on the market.
- Extras: Sun panel 100 W from €120 to 180, 1 kWh station from €500 to 800, Satellite module from €250 to 400.
- The first generation can use an off-the-shelf ARM chip and a separate FPGA card, to reduce the risk of developing the chiplet package.
- Order: public register specification, software emulator, an FPGA prototype of the “2D machine”, 3D on FPGA, the first chiplets at 12 to 28 nm.
- Licences: OS under the GPL, hardware under CERN-OHL-S. Schools, hospitals, co-operatives and public administration without licence fees; commercial manufacturers pay for the “Neo” name and certificate.

Neo · concept 2026 · mascot: Dijana